The macroeconomic effects of gender news: Insights from EU directives

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Abstract

Using a new narrative series based on the national transposition of European Union directives, I identify gender news shocks and study their macroeconomic effects. Gender policy announcements increase female employment and labor-force participation, narrow the employment gap, and raise output over time. While in the aggregate men are not crowded out, sectoral data point to reallocation. Gender news makes women more optimistic on labor market prospects, leading to an increase in participation that lowers female wages. Effects are heterogeneous across policy objectives, transposition timing, and women's characteristics, with work–family reconciliation policies and timely transpositions generating stronger labor-market responses.

JEL classification: E24, E32, C32, J16.

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1 Introduction

Women's growing participation in the labor market has been one of the most important economic and social transformations of recent decades. Yet, despite substantial progress - particularly in advanced economies - gender gaps remain persistent (Olivetti et al., 2025). Closing these gaps has therefore become a central objective on national and international policy agendas, as reflected, for example, in the United Nations Sustainable Development Goals and the European Union's Gender Equality Strategy. But what are the macroeconomic effects of policies aimed at reducing gender disparities? While a vast literature examines the effects of specific gender policies and legal reforms, much less is known about the consequences of broader policy efforts to advance gender equality.

In this paper, I exploit the institutional setting provided by European Union (EU) directives to study the macroeconomic effects of gender-policy news. Gender-equality policy in the EU is often pursued through directives, which are legal acts adopted at the European level and subsequently transposed into national law. A directive is binding on Member States as to the result to be achieved, while leaving national authorities discretion over the form and methods of implementation. Member States must then do so by a specified deadline, generally within two years; failure to comply may trigger infringement proceedings. National transposition thus translates a common European policy objective into a concrete domestic legal commitment.

This two-stage process provides the basis for my narrative approach. I construct a series based on the adoption dates of national legal measures transposing EU directives that promote gender equality. These events reveal future changes in domestic gender policy before they are implemented and therefore capture its anticipated component (Mertens and Ravn, 2012). I then use this series in a panel local-projection framework to trace the dynamic effects of gender news on labor-market outcomes and economic activity across European countries.

The institutional setting helps address the identification challenge of isolating policy movements unrelated to macroeconomic conditions. Both the policy objectives and transposition deadlines are determined at the EU level, limiting the extent to which national legislative action responds to country-specific business-cycle conditions. Moreover, gender-equality policies have an explicitly distributional motivation.[1]

[1]Narrative approaches use official and historical records to classify policy changes according to the motivations underlying their adoption, retaining those judged unrelated to macroeconomic conditions. Distributional objectives are often regarded as plausibly exogenous because they need not respond sys-

Gender news has economically and statistically significant effects on the labor market and economic activity. Announcements of future gender equality policies lead to gradual increases in female labor-force participation and employment, narrowing the employment gap, measured as the ratio of female to male. The delayed responses are consistent with the anticipatory nature of the shock, as announced policies take time to affect outcomes. Female labor-market expectations, however, improve on impact: female unemployment expectations decline by about 0.17 percentage points, preceding the subsequent rise in participation. Wages decline following the shock, suggesting that labor-supply adjustment is an important, though not exclusive, mechanism.

Turning to aggregate outcomes, output increases gradually over time. This expansion is accompanied by a decline in gender-based misallocation, measured by the output gain that would arise if women had the same economic opportunities as men (Goldberg et al., 2025). These positive effects are in line with recent evidence on the macroeconomic gains from reducing gender disparities (Hsieh et al., 2019; Bergholt et al., 2024; Albanesi, 2025).

Do these positive effects come at the expense of men? I find little evidence that men are crowded out in the aggregate data. Male employment does not decline significantly, although male wages fall, possibly reflecting downward wage pressure from the expansion in labor supply. Sectoral data point to reallocation: male employment declines in energy, water, and mining, but increases in other sectors.

Given that EU directives concern different policy targets, I next examine whether their effects vary across these objectives. Work-family reconciliation policies generate larger increases in female labor-force participation and employment than other types of gender-equality measures. I then examine whether the timing of national transposition also matters. Shorter lags between the adoption of an EU directive and its national transposition are associated with stronger employment and wage responses, whereas longer lags primarily increase participation. This pattern suggests that transposition timing may shape the margin through which gender policies affect the labor market, providing policymakers with an additional lever for pursuing specific policy objectives. Finally, the effects also vary across groups of women, according to age, marital status, and educational attainment.

A comprehensive series of sensitivity checks indicates that the results are robust across alternative identification strategies, estimation methods, model specifications, and sam-

tematically to business-cycle developments. For a recent application of this approach, see for instance den Besten et al. (2026).

ple periods. In particular, I obtain similar results when identifying gender-policy news through a maximum forecast error variance approach (Ben Zeev et al., 2025), estimating the effects of the directives using the local-projection difference-in-differences method of Dube et al. (2025), and using the narrative series as an internal instrument in the VAR. The findings also remain broadly unchanged when using annual rather than quarterly data, varying the lag order, considering alternative subsamples, and accounting for the COVID period.

Related literature and contribution. This paper first relates to the literature on gender policies, including Ruhm (1998); Lalive and Zweimüller (2009); Olivetti and Petrongolo (2017); Bertrand et al. (2019); Baker et al. (2023); Kleven et al. (2024); Farré et al. (2026). Most studies examine specific interventions, particularly family policies such as childcare provision and parental leave. I instead study the macroeconomic effects of a broad set of policies aimed at advancing gender equality. To this end, I assemble a new database covering 21 EU gender-equality directives and their national transposition measures across all Member States, including policy objectives, EU adoption and national transposition dates. Exploiting the institutional framework, I provide evidence on the effectiveness of EU directives in pursuing gender-equality objectives.

By focusing on labor-market outcomes and economic activity, this paper also relates to the literature on the causes and macroeconomic consequences of the gender revolution (Greenwood et al., 2005; Ngai and Petrongolo, 2017; Hsieh et al., 2019; Fukui et al., 2023; Bergholt et al., 2024; Albanesi, 2025). Among these contributions, Ngai and Petrongolo (2017) show that structural transformation toward services narrowed gender gaps in hours and wages; Fukui et al. (2023) link the slowdown in business-cycle recoveries to the convergence of female employment toward male employment and find limited crowding out of men; and Bergholt et al. (2024) quantify the aggregate spillovers of gender convergence. I contribute by studying how anticipated gender-equality policies affect labor-market outcomes and economic activity.

From a methodological perspective, this paper builds on the narrative-identification literature, which uses official and historical records to isolate policy changes. This approach has been applied to monetary policy (Romer and Romer, 2004), tax changes (Romer and Romer, 2010; Mertens and Ravn, 2013; Cloyne, 2013), government spending (Ramey, 2011; Ramey and Zubairy, 2018), and, more recently, tariff shocks (den Besten et al., 2026; den Besten and Känzig, 2026). I extend this approach to the identification

of gender-policy news.

Finally, this paper is connected to the literature on news and the macroeconomy, which studies how information about future changes in economic fundamentals or policy affects current expectations and macroeconomic outcomes (Beaudry and Portier, 2006; Jaimovich and Rebelo, 2009; Barsky and Sims, 2011; Schmitt-Grohé and Uribe, 2012; Ben Zeev and Pappa, 2017; Känzig, 2021; Maćkowiak and Wiederholt, 2025). In the spirit of Mertens and Ravn (2012), I use the adoption dates of national legislative measures to identify news about future gender-policy changes and trace its effects.

Outline. The paper proceeds as follows. In Section 2, I discuss the identification design, providing institutional background and details on the construction of the gender news series, and a set of diagnostic tests. Section 3 reports the econometric specification and results. I first examine the main effects of gender-policy news on the labor market and economic activity, before turning to the forward-looking mechanism operating through expectations. I then discuss male responses, sectoral reallocation, and heterogeneity across groups of women, policy types, and transposition lags. In Section 4, I perform a set of sensitivity checks concerning the identification strategy, model specification, and sample period. Section 5 concludes.

2 Identification

Gender-equality policies are often introduced through legislative measures, whose adoption dates and objectives can be recovered from official records. This makes them particularly well suited to narrative identification, which uses historical and institutional information to isolate policy changes (Romer and Romer, 2004, 2010; Ramey, 2011; Mertens and Ravn, 2013; Cloyne, 2013; Ramey and Zubairy, 2018; den Besten et al., 2026). Because legislative adoption generally precedes implementation, these events reveal information about future policy and can therefore be used to construct a narrative series capturing the anticipated component (Mertens and Ravn, 2012) of gender-policies.

A common challenge in economics is to isolate policy movements that are unrelated to prevailing economic conditions. When the analysis relies on the timing of legislative acts, this concern translates into the possibility that the adoption of legislation itself responds to the economic environment. Narrative approaches commonly address this issue by selecting policy documents according to the motivation underlying their adoption,

thereby identifying the policy objective pursued by the legislator. Accounting for it helps distinguish policy changes driven by that objective from those adopted in response to the business cycle. The distributional objective of gender-equality legislation makes a direct cyclical stabilisation motive less salient. However, it does not by itself establish exogeneity, since the timing of adoption may still respond, for example, to labor-market conditions or broader social developments.

To address endogeneity, I focus on the national transposition of EU directives aimed at promoting gender equality, exploiting the institutional framework that governs their transposition. EU directives are legislative acts that set the objectives to be achieved at the EU level and require Member States to adopt corresponding national legislation within specified deadlines. These institutional constraints limit the scope for the legislative acts considered in the analysis to respond to country-specific conditions.

The resulting narrative series is a binary indicator equal to one in periods in which a Member State transposes a directive into national law, and zero otherwise.[2] Before describing its construction in detail, I provide institutional background on the relevant EU legislative framework.

2.1 Institutional background

Equality between women and men is a founding value of the EU and a central objective of its policy agenda. The EU's commitment dates back to the 1957 Treaty of Rome, which introduced the principle of equal pay for work of equal value. Since then, EU action has developed along three main dimensions: equal-treatment legislation, the integration of a gender perspective into all policy areas, and specific measures aimed at advancing women's economic and social position. These objectives are pursued through policy strategies, programs, and legislative actions.

Among the legal instruments available to advance them, the Treaty on the Functioning of the European Union (TFEU) identifies five types of EU legal acts: regulations, directives, decisions, recommendations, and opinions. This work focuses on directives. Their two-stage structure - adoption at the EU level followed by transposition into national law - provides a particularly useful setting for identifying gender-policy news. Moreover, directives have played a central role in the EU's legislative efforts to advance gender equality.

[2]In additional exercises, I also account for intensity (Section 4).

Under Article 288 of the TFEU, a directive is binding for Member States as to the result to be achieved, while leaving national authorities free to choose the form and methods of transposition. Unlike regulations, directives are therefore not generally directly applicable but must be incorporated into domestic law through national transposition measures.

Each directive specifies a transposition deadline, generally within two years of its adoption. By that date, Member States must adopt the measures required to comply with the objective and notify them to the European Commission. Failure to transpose a directive fully or on time may lead the Commission to initiate infringement proceedings.[3] Thus, although national authorities retain discretion over the legal instruments used, both the policy objective and the implementation window are determined at the EU level. This structure motivates the use of national transpositions to construct the gender-policy news series, as described in the next section.

2.2 Construction of the series

The information used to construct the gender news series is drawn from EUR-Lex, the official online portal providing access to EU law and legal documents. To identify gender-related measures, I first retrieve all EU directives that contain the terms "gender" and/or "women" in either their title or text. I then review their content individually and retain only directives whose objectives are directly related to gender equality. This screening procedure yields a final sample of 21 directives.

The selected acts cover a broad range of policy areas. Examples include Directive (EU) 2022/2381 on improving gender balance among directors of listed companies; Directive 2006/54/EC on equal opportunities and equal treatment in employment and occupation; Directive (EU) 2024/1385 on combating violence against women and domestic violence; and Directive (EU) 2019/1158 on work–life balance for parents and carers.[4]

For each selected directive, I collect from EUR-Lex the national measures adopted by

[3]Infringement proceedings are a common enforcement mechanism. They involve several stages: a letter of formal notice, a reasoned opinion, a possible referral to the Court of Justice of the EU, and, ultimately, potential financial sanctions. To give an order of magnitude, in 2023 the Commission closed 1,038 infringement cases, 94% of which were resolved without a decision to refer the Member State to the Court of Justice. Late transposition of directives is among the most frequent grounds for initiating such proceedings.

[4]While the paper's main objective is to provide evidence on the effects of gender policies overall, different policy objectives may have different effects. Section 3.5 therefore examines whether the effects vary across directive categories.

Member States to transpose it into domestic law and their corresponding adoption dates. Member States may also notify pre-existing legislation that already satisfies the directive's requirements. I exclude these measures and retain only national acts adopted after the corresponding EU directive, thus ensuring that the series captures new legislative action following the adoption of the directive rather than legislation already in force.

I then construct a country-specific indicator equal to one in a period in which at least one qualifying national transposition is adopted and zero otherwise. The baseline series is at quarterly frequency, although the underlying daily dates allow it to be constructed at alternative frequencies.

Figure 1 shows the resulting indicator for the countries included in the baseline sample. The series contains 29 national transposition events for France, 17 for Greece, 9 for Italy, and 39 for Portugal.[5]

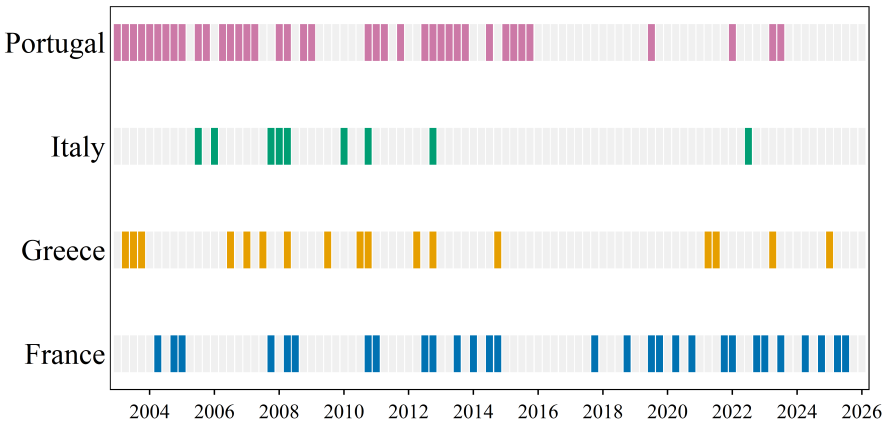


Figure 1: Gender news series by country

Notes: Each row represents a country and each cell a quarter. Colored cells identify quarters in which the gender-policy news indicator equals one, corresponding to the national transposition of an EU gender-equality directive.

2.3 Diagnostics of the series

I begin by discussing the information conveyed at the national transposition stage, relative to the earlier adoption of the corresponding EU directive. I then conduct a set of exercises to assess the validity of the series. Specifically, I test for the predictability of transposition delays and for pre-existing trends in the outcomes; examine whether transpositions can

[5]Although the baseline analysis focuses on these four countries because of labor market data availability, I construct the gender-policy news series for all EU Member States. Figure B.1 in the Appendix reports the complete series.

be predicted by past macroeconomic and labor-market conditions; conduct a placebo exercise; and assess whether national transpositions capture public attention.

Information content of national transpositions. Directives establish policy objectives at the EU level, which Member States subsequently transpose into national law at different points in time. The relevant policy sequence therefore comprises three stages: the adoption of the EU directive, its national transposition, and the date on which the resulting policy becomes effective (Figure 2).

A natural question is why the analysis focuses on the second step. The adoption of an EU directive communicates a common policy objective and may already affect expectations. However, national transposition specifies how this objective is incorporated into domestic law, thereby reducing uncertainty about the content, applicability, and effective domestic commitment of the policy.[6] For these reasons, national transposition is a plausible relevant information date for households, firms, and investors. The two exercises below provide further evidence by comparing financial-market reactions around national transpositions and the adoption of EU directives.[7]

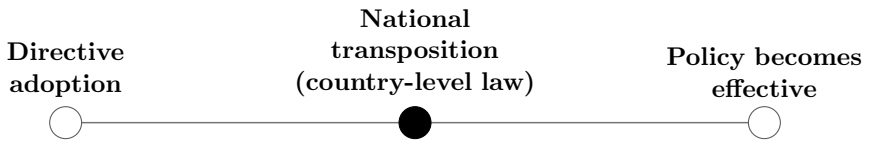


Figure 2: Institutional process timeline

The first exercise compares stock-price movements around the two types of events for firms particularly exposed to gender-equality developments. Its design combines the firm-selection strategy of Fisher and Peters (2010) with the use of asset-price changes commonly employed for high-frequency identification in the monetary-policy, oil, and, more recently, fiscal-policy literature (Gürkaynak et al., 2005; Känzig, 2021; Endo et al., 2026). Following Fisher and Peters (2010), who identify defense firms using a ranking of military contractors and examine their stock returns, I select firms plausibly more exposed to gender-equality developments by ranking listed companies according to their

[6]Figure B.2 in the Appendix reports the response of the Economic Policy Uncertainty index (Baker et al., 2016) to a national transposition and shows that it declines following this event, though not significantly.

[7]Focusing on national transposition rather than the effective date also allows the analysis to isolate the anticipated component of gender policies. In the spirit of Mertens and Ravn (2012), who distinguish anticipated from unanticipated tax changes using the implementation lag between a measure becoming law and taking effect, national transpositions are treated as news about future changes in domestic gender policy.

female board shares, obtained from LSEG Datastream. I retain the four firms with the highest female board shares in each country.[8] For each firm and event date, I compute the daily stock-price change as $\Delta P_{i,t} = P_{i,t} - P_{i,t-1}$, and compare country-level average absolute changes around EU directive adoptions and national transpositions. Stock-price movements are larger around national transpositions: the average absolute price change is approximately 16 percent greater than that observed around the adoption of EU directives. This larger market reaction is consistent with national transpositions being more salient information events for investors than the earlier adoption of EU directives.

As a second exercise, I repeat the comparison using market-adjusted abnormal returns rather than raw stock-price changes. This approach is closer to the excess-return methodology of Fisher and Peters (2010) and removes movements common to the European equity market. Specifically, for each firm i and event day t, I compute the excess returns relative to the STOXX600 index as $AR_{i,t} = R_{i,t} - R_t^{\text{STOXX600}}$. I then take the absolute value of daily abnormal returns and average it across firms and events within each event type. This sample-level mean absolute abnormal return measures the extent to which firms' returns depart from aggregate market movements around EU directive adoptions and national transpositions. Abnormal-return movements are larger around national transpositions: the sample-level mean absolute abnormal return is approximately 22.5 percent higher than around EU directive adoptions.[9] This result reinforces the evidence that national transpositions are salient.

As an additional illustrative example, I examine media attention to the two types of events, using Google News search results as a proxy for their prominence in the press. For this exercise, I focus on Spain and compare the adoption of Directive (EU) 2022/2381, which aims to improve gender balance among directors of listed companies, with its national transposition, *Ley Orgánica 2/2024, de 1 de agosto, de representación paritaria y presencia equilibrada de mujeres y hombres*. Specifically, I searched Google News for *Directiva 2022/2381* and *Ley Orgánica 2/2024*, with Spanish selected as the search language and Spain as the country. As reported in Table 1, the searches returned 10 and 50 results, respectively. This difference suggests that the national transposition received more media attention than the adoption of the directive.[10]

[8] When fewer than four eligible firms are available, I retain only the highest-ranked firm.

[9] This comparison includes all countries for which data on firms exposed to gender-equality developments are available. Restricting the sample to the four countries included in the baseline macroeconomic analysis, the corresponding difference is 18.25 percent.

[10] This exercise is relevant in light of the forward-looking mechanism proposed below, according to which gender news may affect macroeconomic outcomes by shaping expectations.

Table 1: Google News coverage of the directive and its Spanish transposition

Search	Number of results
Directive 2022/2381	10
Ley Orgánica 2/2024	50

Note: Google News search results for Directive (EU) 2022/2381 in Spain and its Spanish legal transposition.

Predictability of transposition timing. Despite the EU-imposed transposition deadline, national transposition timing may still reflect pre-existing economic and labor-market conditions. I therefore test whether the delay between a directive's adoption and its national transposition is predicted by lagged GDP, female employment, female labor-force participation, and the female-to-male employment ratio. Table 2 reports regressions including country and directive fixed effects. None of these variables significantly predicts transposition delays, providing no evidence that transposition timing systematically responds to observable conditions in the outcomes of interest.

Table 2: Predictability of transposition timing

	Delay in months
GDP	−0.034
	(0.538)
Female employment	0.907
	(0.453)
Female labor-force participation	0.815
	(0.472)
Female-to-male employment ratio	−0.849
	(1.060)

Notes: The table reports estimates from a regression of the delay between the date of an EU directive and the date of its national transposition. The delay is measured in months. The explanatory variables are measured in the quarter preceding the national transposition and are included jointly in the regression. The specification includes country and directive fixed effects. Standard errors are clustered at the country level and reported in parentheses.

Absence of pretrends. A potential concern is that national transpositions may occur after systematic changes in macroeconomic or labor-market conditions have already begun. Such pre-existing dynamics would make it difficult to distinguish the effects of gender-policy news from the continuation of earlier trends. To assess this possibility, I estimate an event-study specification around national transposition dates. Figure 3 reports

the estimated responses of GDP, the employment ratio, and the wage ratio before the event. The coefficients preceding national transpositions are close to zero and statistically insignificant, providing no evidence of systematic pre-trends.

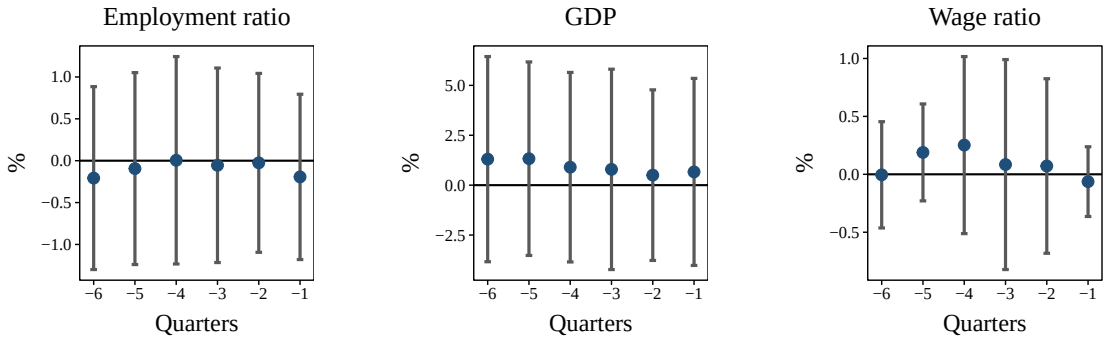


Figure 3: Pre-trends before national transpositions

Notes: The figure reports the estimated coefficients for the quarters preceding a national transposition. The dots denote the point estimates, and the vertical bars represent 95 percent confidence intervals.

Predictability of transposition events. To check whether macroeconomic and labor market inequality variables have any power in forecasting the gender-policy news series, I estimate the following regression:

$$z_{i,t} = \alpha_i + \lambda_t + \sum_{\ell=1}^{6} \left(\beta_\ell^G \mathrm{GDP}_{i,t-\ell} + \beta_\ell^E \mathrm{ER}_{i,t-\ell} + \beta_\ell^W \mathrm{WR}_{i,t-\ell} \right) + \varepsilon_{i,t}. \tag{1}$$

Here, $z_{i,t}$ is the national transposition indicator, α_i and λ_t denote country and time fixed effects, and the regressors include six lags of GDP, the female-to-male employment ratio (ER), and the female-to-male wage ratio (WR). Standard errors are clustered at the country level. Table 3 reports the results. The coefficients are generally imprecisely estimated and display no systematic pattern across variables or lags. Overall, the evidence provides little indication that national transpositions can be predicted by past macroeconomic conditions or inequalities in the labor market.

As an alternative exercise, and following the approach of Ghomi and Pappa (2026), I estimate a pooled probit model in which the probability of a national transposition is modeled as a function of lagged GDP, the female-to-male employment ratio, and the female-to-male wage ratio. The results are unchanged: past macroeconomic conditions and labor-market gender gaps do not significantly increase the probability of a national

transposition.[11]

Table 3: Predictability of national transpositions

	$t-1$	$t-2$	$t-3$	$t-4$	$t-5$	$t-6$
GDP	−0.025	−0.009	−0.026	−0.092	0.071	0.042
	(0.043)	(0.019)	(0.012)	(0.030)	(0.024)	(0.018)
Employment ratio	−0.014	−0.027	0.155	−0.538	0.661	−0.369
	(0.194)	(0.311)	(0.211)	(0.437)	(0.405)	(0.158)
Wage ratio	−0.028	0.108	−0.159	0.158	−0.082	0.0002
	(0.048)	(0.105)	(0.098)	(0.108)	(0.090)	(0.050)

Notes: The table reports estimates from the predictability regression. The dependent variable is an indicator equal to one when a national transposition occurs and zero otherwise. Each column reports the coefficient associated with the indicated lag of GDP, the female-to-male employment ratio, and the female-to-male wage ratio. All lags are included jointly in the regression, together with country and time fixed effects. Standard errors clustered at the country level are reported in parentheses.

Placebo test. To assess whether the baseline results could arise from coincidental timing or broader trends, I conduct a placebo exercise. Specifically, I construct a fictitious gender-news series by randomly reallocating, within each country, the same number of national transposition events observed in the data across the available quarters. I then re-estimate the baseline local projections using this placebo series. Figure B.3 in the Appendix reports the results. As expected for a placebo treatment, the estimated responses are not statistically significant.

Public attention around national transpositions. To assess whether national transpositions attract public attention, I examine Google Trends search activity around the dates on which they are adopted as national legal acts. Google Trends reports relative search interest for a given term over a selected period on a scale from 0 to 100. Higher values denote greater interest: 100 corresponds to the peak level of interest in the displayed period, 50 to half of that peak, and zero indicates insufficient data.

Figure 4 presents two illustrative examples. The upper panel shows search interest in France for the term *quotas* around the national transposition, on 16 October 2024, of Directive (EU) 2022/2381 on improving gender balance among directors of listed companies. The lower panel reports search interest in Portugal for *licença parental* around

[11] To keep the discussion concise, I do not report the estimates here; they are available upon request.

the national transposition, on 10 January 2022, of Directive (EU) 2019/1158 on work–life balance for parents and carers. In both cases, searches for terms closely related to the content of the legislation increase around the adoption of the national legal act. While these descriptive patterns do not establish a causal effect, they provide suggestive evidence that national transpositions are events that attract public attention.

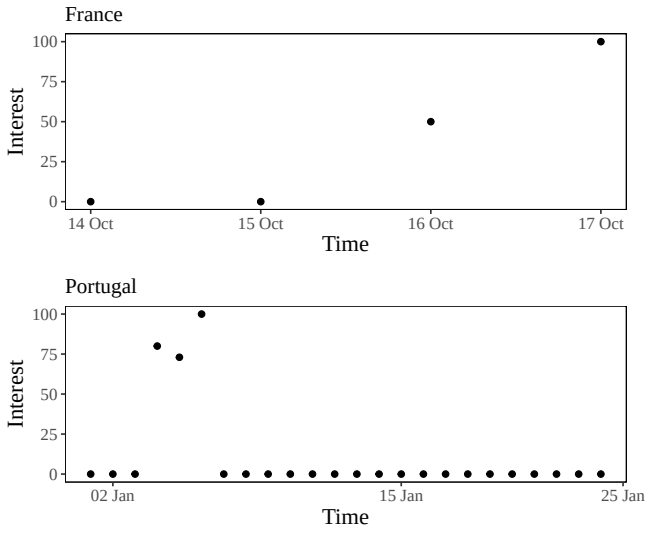


Figure 4: Google Trends interest around national transpositions

Notes: Google Trends relative search interest, normalized from 0 to 100. The upper panel reports searches for *quotas* in France around the transposition of Directive (EU) 2022/2381 (16 October 2024); the lower panel reports searches for *licença parental* in Portugal around the transposition of Directive (EU) 2019/1158 (10 January 2022). Source: Google Trends.

3 The macroeconomic effects of gender news

3.1 Econometric specification

To assess whether gender news affects the labor market and the broader economy, I estimate its dynamic effects using local projections (Jordà, 2005) in a panel setting. For each horizon h, I estimate the following regression:

$$y_{i,t+h} = \alpha_i^h + \beta_h z_{i,t} + \sum_{p=1}^{4} \Gamma_{h,p} x_{i,t-p} + \varepsilon_{i,t+h}, \tag{2}$$

where $y_{i,t+h}$ denotes the outcome of interest. The baseline outcomes are the female-to-male employment and wage ratios, female employment, wages, and labor-force partici-

pation. The variable $z_{i,t}$ denotes the gender news shock, while $x_{i,t-p}$ includes four lags of all baseline outcomes, and α_i^h are country fixed effects. The coefficient β_h traces the response of the outcome at horizon h to a gender news shock.

The baseline sample is an unbalanced quarterly panel spanning the period from 2005Q1 to 2026Q1 at most, with the exact coverage varying across countries and outcomes. All impulse responses represent a one-unit increase in the shock, corresponding to one additional national transposition. Standard errors are computed using the Driscoll–Kraay covariance estimator, which allows for heteroskedasticity, serial correlation, and cross-sectional dependence.

Section 4 assesses the sensitivity of the results to alternative identification and estimation approaches, as well as to changes in the model specification and estimation sample.

Data. The baseline labor-market variables are drawn primarily from the International Labor Organization's database (ILOSTAT) and Eurostat. The additional exercises use microdata from the European Union Statistics on Income and Living Conditions (EU-SILC) and from the European Central Bank's Consumer Expectations Survey (CES), and the Global Gender Distortion Index (GGDI) constructed by Goldberg et al. (2025). A more detailed description of the data sources, variable definitions, and transformations is provided in Appendix A.

3.2 Effects on the labor market and the macroeconomy

Figure 5 reports the baseline impulse responses to a positive unit gender news shock, corresponding to one national transposition. The solid black lines denote the point estimates, while the dark and light shaded areas represent the 68 and 90 percent confidence bands, respectively.

A positive gender news shock narrows the gender employment gap, with the female-to-male employment ratio increasing by about 0.2 percent at its peak after one year. This response is driven by a persistent increase in female employment. Female wages decline, while female labor-force participation increases. This joint pattern points to an expansion in female labor supply. This finding is consistent with Hyland et al. (2020), who show that reforms strengthening women's legal rights are associated with greater female labor-force participation.

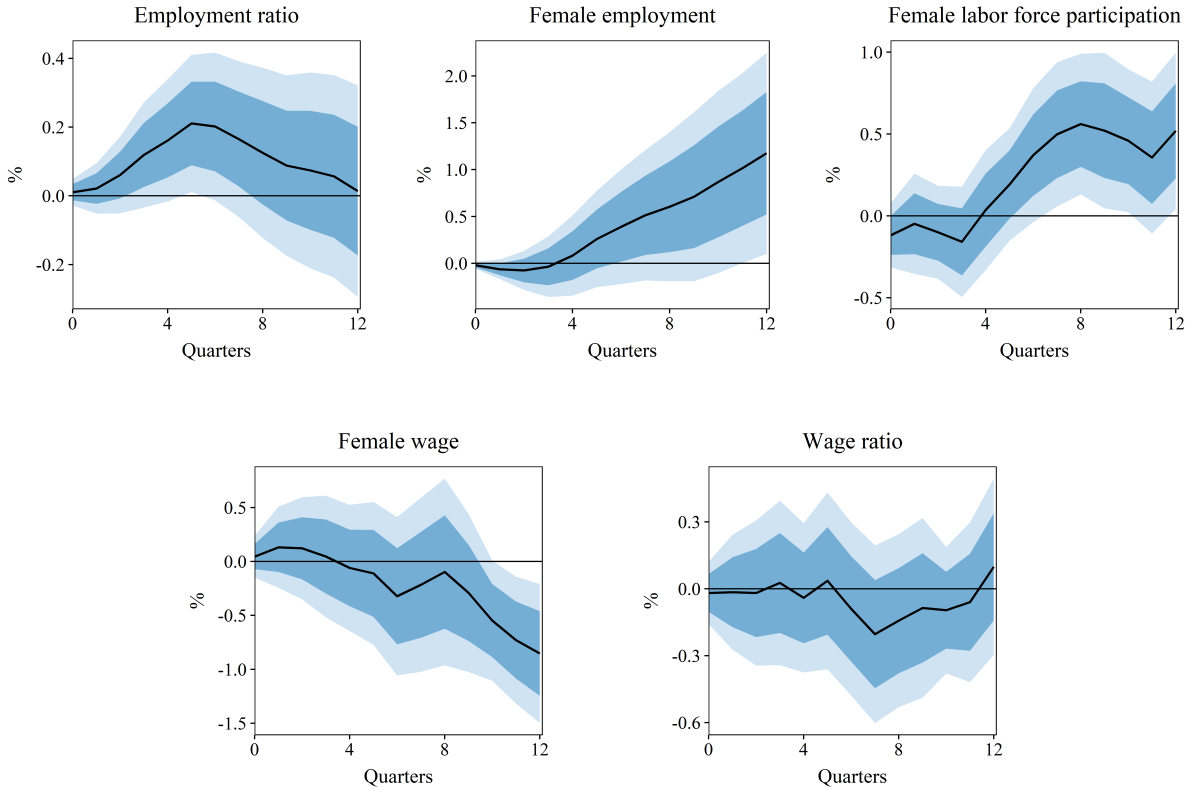


Figure 5: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

All responses are statistically indistinguishable from zero on impact and become significant only at longer horizons, in line with the anticipatory nature of the shock. The female-to-male wage ratio remains broadly unchanged, suggesting that the resulting wage pressure is not specific to female workers.

Gender news appears to have the intended effect of reducing labor-market inequalities, at least along the employment margin, while encouraging female participation. A natural question is whether these labor-market gains have broader macroeconomic effects and, if so, in which direction. To address this, I augment the baseline specification with real gross domestic product (GDP). Figure 6 reports the result. Output does not respond on impact but increases gradually, with the response becoming significant only after three years. This delayed expansion is consistent with recent evidence on the positive macroeconomic effects of the gender revolution (Bergholt et al., 2024).

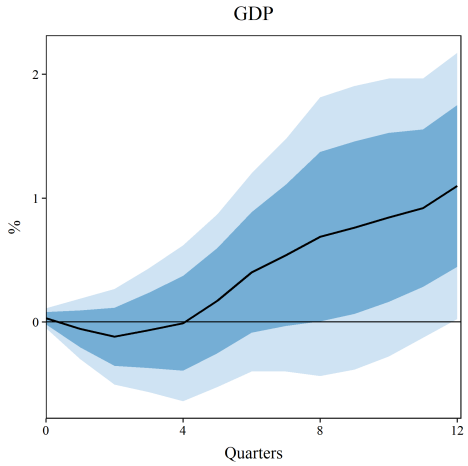


Figure 6: Impulse responses of GDP after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

One explanation proposed in the literature for these positive effects on economic activity is a reduction in misallocation. By weakening the distortions that prevent women from accessing employment opportunities, gender-equality policies may improve the allocation of talent and raise aggregate outcomes (Hsieh et al., 2019). To examine this aspect, I estimate the response of the Global Gender Distortion Index constructed by Goldberg et al. (2025). It measures the aggregate productivity losses associated with gender-based distortions - that is, how much higher GDP would be if women faced the same opportunities as men. Figure 7 shows that the index declines following the shock, becoming significant only at longer horizons and reaching its largest reduction in the eleventh quarter. This response is consistent with gender news raising output partly through a gradual reduction in gender-based misallocation.

To shed further light on the macroeconomic effects, I examine the response of inflation. Figure 8 reports the result. Inflation does not respond significantly, although its estimated response is negative. Together with the increase in female employment and labor-force participation, this pattern might be consistent with a labor-supply adjustment and does not point to a purely demand-driven expansion.

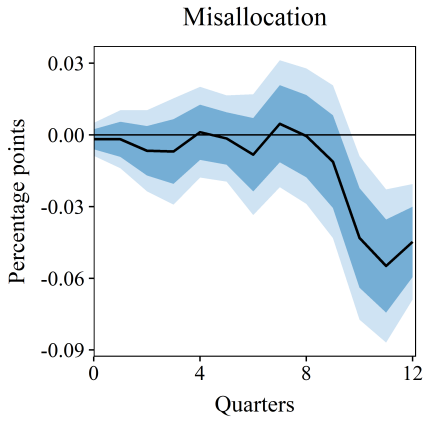


Figure 7: Impulse responses of the global gender distortion index after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

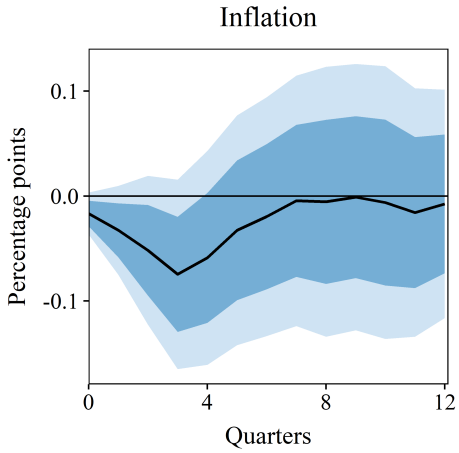


Figure 8: Impulse responses after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

3.3 Propagation channel

To shed light on the mechanisms through which gender news propagates and generates labor-supply responses, I examine whether these shocks shape women's expectations about future labor-market conditions.

The proposed mechanism is inherently forward-looking. In dynamic models of female labor supply, participation depends on the expected value of labor-market attachment rel-

ative to non-participation (Attanasio et al., 2008). A national transposition may therefore affect behavior before its provisions are fully implemented if it provides credible information that the future labor market will become more favorable to women. Depending on the directive, this may reflect lower expected discrimination, stronger legal protection, or better compatibility between work and family life.



Figure 9: Impulse responses of female unemployment expectations after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

Figure 9 provides evidence consistent with this channel. Following gender-equality news, women become significantly less pessimistic about future aggregate unemployment on impact, before female participation and employment begin to increase gradually. This timing is consistent with women revising their perceived labor-market prospects and subsequently adjusting their participation decisions.[12]

The proposed forward-looking adjustment is compatible with a discouraged-worker mechanism, whereby more favorable perceived job-finding prospects increase the expected returns to labor-market entry (Dagsvik et al., 2013). It can also be interpreted in light of evidence that beliefs about the returns to maternal employment shape intended labor-supply choices (Boneva et al., 2026). More broadly, the result relates to recent work by Reiche and Maffei-Faccioli (2026), which highlights the macroeconomic relevance of women's perceptions. They show that women and men revise their labor-market expectations differently in response to common inflationary shocks, with these divergent beliefs translating into gender-differentiated wage outcomes.

[12]Figure B.4 reports the corresponding results for male unemployment expectations, which show no significant response to the gender news shock.

A possible related concern is that national transpositions may be accompanied by fiscal measures, so that the observed baseline responses could partly reflect changes in public expenditure. To assess this possibility, Appendix Figure B.5 reports the response of government expenditure to gender news. Expenditure does not increase on impact and responds only at longer horizons. This pattern is consistent with the interpretation that the immediate improvement in women's labor-market expectations is not driven by a contemporaneous fiscal expansion, although subsequent public spending may contribute to the medium-run effects of the policies.

3.4 Wider effects

I then consider a broader set of outcomes, examining potential spillovers to male workers, adjustments along the intensive margin, labor-demand responses, productivity, and sectoral reallocation.

Male responses. Gender news appears to generate positive macroeconomic effects. Do these gains come at the expense of men? The aggregate responses of male employment and wages provide little evidence of such a trade-off. Figure 10 shows that male employment declines slightly on impact and subsequently increases, with the response remaining statistically insignificant throughout. Thus, the rise in female employment does not appear to crowd out male employment at the aggregate level. Consistent with this finding, the additional results reported in Appendix, Figures B.6 and B.7, show no significant response of either aggregate or male unemployment.

Male wages decline at longer horizons. This adjustment suggests that the expansion in female labor supply exerts broader downward pressure on wages.[13] Moreover, the limited degree of crowding out echoes the findings of Fukui et al. (2023), who find that the long-run expansion of female employment generated only modest displacement of male employment.

Intensive margins. The results so far concern the extensive margin. I now turn to the intensive margin by examining female hours worked. Figure 11 shows that hours worked increase following a gender-news shock. Together with the rise in female employment,

[13]Related evidence is provided by Acemoglu et al. (2004), who use cross-state variation in male mobilization during World War II as a source of plausibly exogenous variation in female labor supply and find that an increase in female labor supply reduces wages.

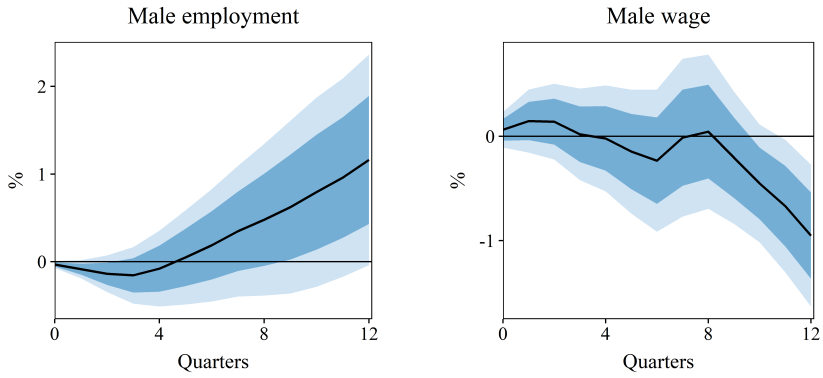


Figure 10: Impulse responses of male employment and wages after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

this points to an increase in female labor input. Figure B.8 in the Appendix shows that female part-time employment also increases, suggesting that some of the employment gains take the form of part-time jobs. This pattern is consistent with the inclusion of work–family reconciliation measures among the policies considered, as such measures aim to make paid work more compatible with family responsibilities.

These patterns also qualify the interpretation of the decline in average female wages. The fall may reflect an equilibrium response to an outward shift in female labor supply, while also reflecting composition if the policy induces entry among women with relatively lower potential wages or into part-time jobs. Both explanations are consistent with the observed joint dynamics.

Labor demand. The evidence so far points to labor supply as an important margin of adjustment. However, firms may also respond to gender-policy news by increasing their demand for labor. To evaluate the presence of this channel, I estimate the baseline local-projection specification using the number of job vacancies as the outcome variable. Figure 12 reports the results. Vacancies increase, if anything, but the estimated responses are imprecise and generally not significant. Thus, the evidence provides limited support for a labor-demand channel and is consistent with labor supply being the main margin of adjustment after news concerning future implementation of gender policies.

Productivity. The positive response of GDP and the decline in misallocation raise the question of whether a larger female presence in the labor market translates into higher

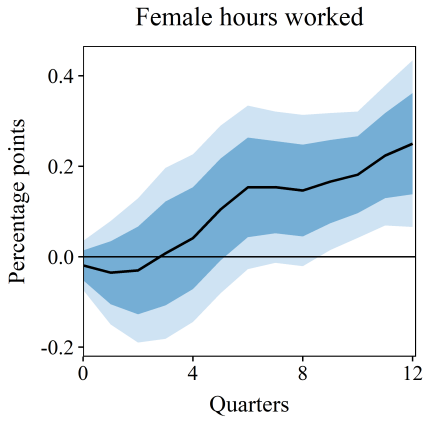


Figure 11: Impulse responses of female hours worked after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

productivity. Figure B.9 in the Appendix reports the response of labor productivity following a gender-news shock. The evidence is not conclusive: productivity increases slightly at very short horizons, but the response is not significant and subsequently fades. This pattern suggests that the aggregate output effects of gender-policy news initially operate mainly through higher employment and labor-force participation, rather than through an immediate increase in output per worker. Productivity gains associated with a more efficient allocation of talent may instead take longer to materialize and are difficult to detect in the sample.[14]

Sectoral analysis. The absence of aggregate male crowding out does not rule out adjustments across sectors. A subsequent question is therefore whether gender news changes the sectoral allocation of employment, with workers moving away from some industries and toward others. To examine this possibility, I estimate the responses of male employment across sectors. Figure 13 shows the results.

Sectoral responses point to a reallocation of male employment across industries. Employment declines in energy, water, and mining, while it rises in other sectors, including agriculture and manufacturing. Thus, the fall in gender inequality partly reflects weaker male employment in some sectors, alongside gains in others, rather than a broad-based decline in male employment. This pattern relates to a broader literature linking gender

[14]Since a direct measure of productivity is not available at the quarterly frequency for all countries in the sample, I proxy labor productivity by real GDP over employment.



Figure 12: Impulse responses of vacancies after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

gaps to sectoral reallocation (Ngai et al., 2024).

The displacement of male employment in some sectors is consistent with men and women being imperfect substitutes in production. An expansion in female labor supply may therefore reduce the demand for male workers in particular sectors or worker groups, without implying aggregate crowding out. The extent of substitutability is likely to depend on workers' characteristics, including education and age, as well as on sector-specific production technologies. Consistent with this interpretation, Acemoglu et al. (2004) find that the expansion of female labor supply in the United States reduced the wages of both women and less-educated men, suggesting that women were relatively closer substitutes for men with a high-school education. Exploring these margins of substitution, and their role in shaping sectoral reallocation following gender-policy news, is an interesting direction for future research.

3.5 Heterogeneous effects across women, policy type, and transposition timing

This section examines whether the effects of gender-policy news vary with the type of policy instrument, the timing of national transposition, and women's characteristics.

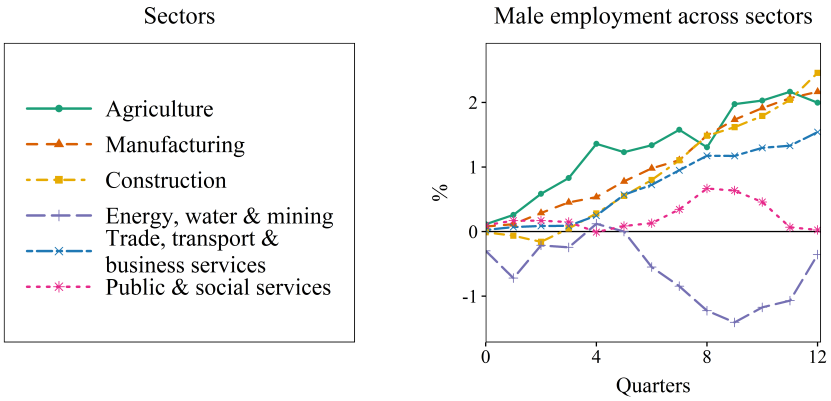


Figure 13: Sectoral responses of male employment to a gender news shock

Notes: The figure reports impulse responses of male employment across six sectors to a unit gender news shock, corresponding to one national transposition. Lines denote point estimates from panel local projections. Colors, line types, and markers identify sectors.

3.5.1 Heterogeneity by policy type

The analysis so far has treated all directives promoting gender equality as a single policy category. However, different policy instruments may affect labor-market outcomes with varying intensity and through distinct channels. To examine this heterogeneity, I classify the directives according to their main policy objective and estimate their effects separately. Figure 14 reports the responses to directives concerning work–family reconciliation and equal treatment and anti-discrimination.

Work–family reconciliation policies generate larger responses than the aggregate gender-policy news shock. In particular, they produce a stronger and more persistent increase in female labor-force participation. This finding is consistent with Olivetti and Petrongolo (2017), who document that family policies facilitating the combination of work and motherhood - particularly childcare and early-education provision - are associated with higher female labor-force participation and narrower gender gaps. Equal-treatment and anti-discrimination directives broadly reproduce the dynamics of the aggregate shock but generate more pronounced wage responses.

Work-family reconciliation policies may be particularly effective for prime-age women, who are more likely to face childcare and other family responsibilities that constrain labor-force participation. To assess this, Figure 15 compares the response of prime-age women's labor-force participation to work-family reconciliation policies with its response to the aggregate gender shock. The increase in participation is stronger and more persistent following work-family reconciliation policies, supporting the interpretation that these

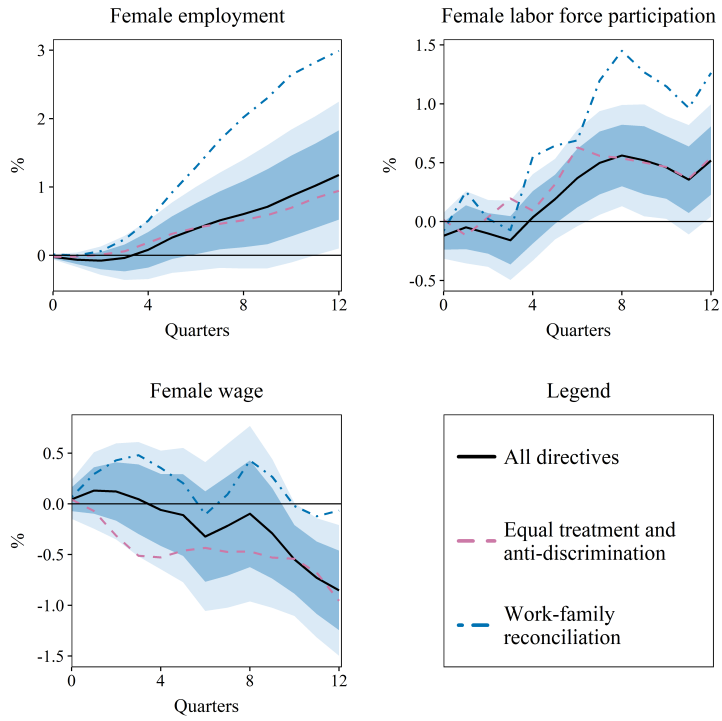


Figure 14: Impulse responses by type of gender-policy directive

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. Each colored line represents a distinct gender-policy news shock, classified according to the policy objective of the underlying directive. The solid lines denote the point estimates, while the dark and light shaded areas represent the 68 and 90 percent confidence bands, respectively.

measures help relax barriers to labor-market attachment for women at ages when family responsibilities are most likely to be binding.

3.5.2 Implementation timing

The time taken by Member States to transpose an EU directive into national law may shape its labor-market effects. A shorter transposition lag may reduce policy uncertainty and signal a stronger commitment to the directive's objectives, allowing workers and firms to adjust. To investigate this dimension, I distinguish between directives transposed within one year of their adoption and those transposed after more than one year. Figure 16 reports the results. Shorter transposition lags generate stronger employment and wage responses, whereas longer lags operate mainly through increased female labor-force participation. These findings suggest that timely transposition may strengthen employment gains, while longer transposition processes are associated primarily with adjustments along the participation margin.

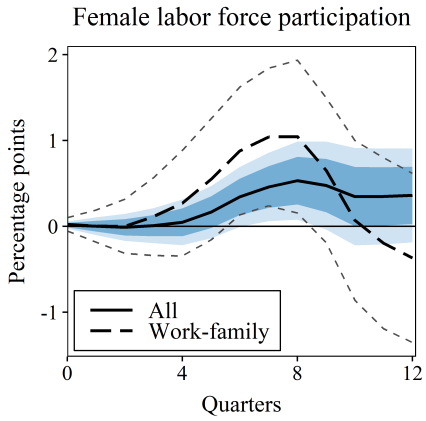


Figure 15: Impulse responses of prime-age female labor-force participation to aggregate gender-policy news and work-family reconciliation policies

Notes: The figure reports impulse responses of prime-age women (aged 25-54) to a unit shock, corresponding to one national transposition. It compares the responses to all gender-policy announcements with those to work-family reconciliation policies.

3.5.3 Heterogeneity across women

Age. Different groups of women may respond differently to gender-policy news. I first examine heterogeneity by age, as women at different stages of their lives and careers may be differentially affected by policies that facilitate, for instance, the reconciliation of work and family responsibilities. I distinguish between prime-age women (aged 25-54) and older women (aged 55-64). Figure 17 reports the impulse responses of labor-force participation for the two groups. Prime-age women display a pattern similar to the baseline result: their labor force participation increases significantly at longer horizons. By contrast, the labor force participation of older women does not appear to respond to gender-policy news.[15]

Marital status. The effects of gender-policy news may also vary by marital status. Figure B.10 in the Appendix reports the responses of married and single women. Consistent with the idea that policies promoting gender equality can relax constraints on labor-market participation, the response of married women is stronger than that of single women. Married women may face greater family-related constraints, so that policies improving the compatibility of work and family life generate a larger increase in their participation. By contrast, single women may face fewer such constraints and already

[15]Data for this exercise are available only for Italy and Portugal; the estimation sample is therefore shorter than in the baseline analysis.

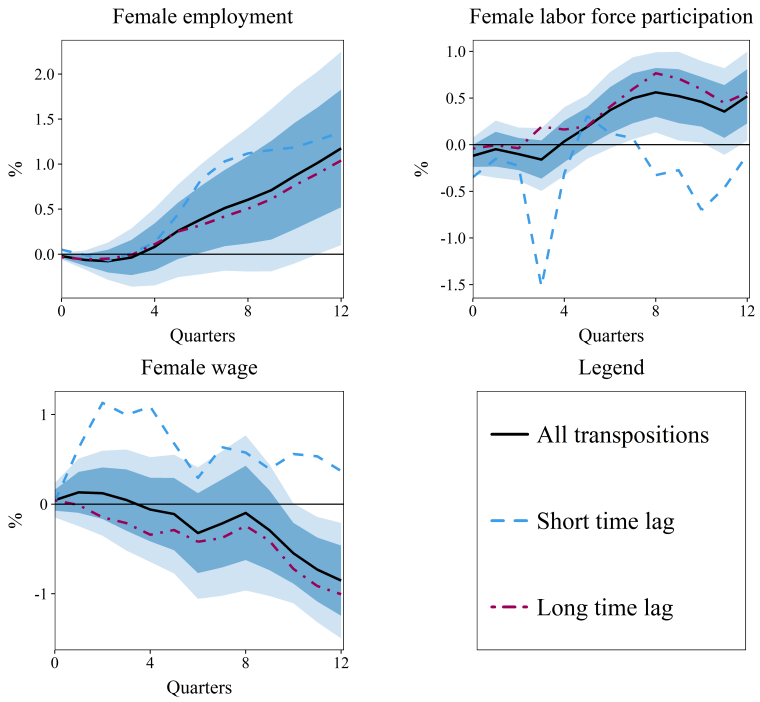


Figure 16: Impulse responses of gender news by transposition timing

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. Each colored line represents a distinct gender-policy news shock, classified according to the timing of transposition. The solid lines denote the point estimates, while the dark and light shaded areas represent the 68 and 90 percent confidence bands, respectively.

participate more extensively in the labor market, leaving less scope for an additional response.

Education. The effects of gender-policy news may vary with educational attainment, which proxies for differences in skills, labor-market opportunities, and the constraints women face. I therefore distinguish between women with basic and advanced education. Figure B.11 in the Appendix reports the results. The estimates suggest that gender-policy news may generate somewhat stronger improvements among women with advanced education, whereas the responses of women with basic education are smaller and less systematic. Consistent with this interpretation, the female-to-male wage ratio for women with advanced education increases significantly following policy news. These results suggest that gender-policy announcements may be particularly effective at improving relative labor-market outcomes among more highly educated women.

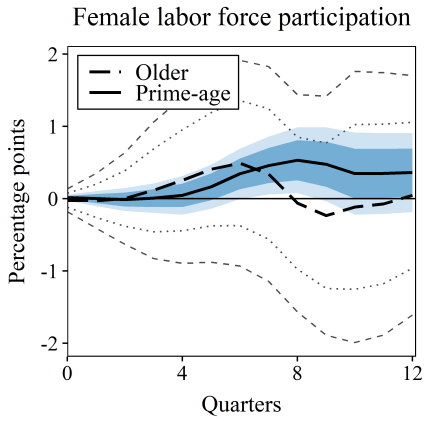


Figure 17: Impulse responses of female labor-force participation to a gender news shock, by age group

Notes: The figure reports impulse responses to a unit gender news shock, corresponding to one national transposition, for prime-age and older women. The solid and dashed lines denote the point estimates for the two age groups; the corresponding dark and light shaded areas indicate 68 and 90 percent confidence bands, respectively.

4 Sensitivity analysis

In this section, I assess the robustness of the baseline results along several dimensions. In particular, I consider alternative identification strategies, conduct additional identification tests, and examine the sensitivity of the results to changes in the model specification and estimation sample. The corresponding results are reported in Appendix.

4.1 Identification

Maximum forecast error variance VAR. As an alternative to the narrative identification, I identify gender news using a VAR-based maximum-share approach (Ben Zeev and Pappa, 2017; Ben Zeev et al., 2025). Specifically, the gender news shock is defined as the innovation that maximizes the forecast error variance of the World Bank's Women, Business and the Law Index over a horizon of six quarters, subject to having no contemporaneous effect on the index.[16] The index measures the extent to which a country's laws provide women with the same economic rights and opportunities as men, covering areas such as employment, pay, marriage, parenthood, entrepreneurship, assets, and pensions. Higher values indicate greater gender equality. Because the index is based on legislation,

[16]This restriction reflects the slow-moving nature of gender equality: policy news may affect expectations immediately, whereas measurable changes in gender equality are likely to materialize only gradually.

it closely matches the legal nature of the policy measure used in the baseline analysis.[17]

Figure B.12 reports the results. Following the gender news shock, the gender employment gap narrows, while female employment and labor-force participation increase. The responses of the wage variables are not significant. Overall, this alternative identification broadly confirms the baseline findings.

Lp-DiD. As a further alternative, I use the local-projection difference-in-differences approach recently proposed by Dube et al. (2025). This method is well suited to the present setting because it estimates dynamic treatment effects by comparing changes in outcomes around national transpositions with those observed for clean control observations that are not exposed to nearby policy events. The gender-policy news dummy is therefore used as an indicator for the occurrence of a national transposition.

Figure B.13 reports the results. Following a national transposition, the gender employment gap narrows and female labor-force participation increases. Unlike in the baseline and previous exercises, the wage ratio also responds positively, indicating an improvement in women's wages relative to men's. Overall, the lp-DiD estimates reinforce the conclusion that gender-policy measures reduce labor-market inequalities.

4.2 Model specification and sample

Recursive VAR. To assess whether the results are robust to the estimation method, I replace the local projections with a recursive VAR. The gender-policy news series is included as the first variable in the recursive ordering, thereby serving as an internal instrument. Figure B.14 reports the resulting impulse responses. Consistent with the baseline estimates, gender-policy news narrows the gender employment gap and increases female labor-force participation. These effects emerge gradually, in line with the anticipatory nature of the shock.

Treatment intensity and policy accumulation. To assess whether the baseline results are sensitive to measuring gender-policy announcements as a binary treatment, I account for intensity. Specifically, I consider the number of national transpositions occurring in each country-quarter and construct a cumulative stock of directives transposed

[17]The index is available at an annual frequency. To conduct the quarterly analysis, I assign each annual observation to all four quarters of the corresponding year.

before that quarter. I include this stock among the lagged controls in the baseline local-projection specification. Figure B.15 in the Appendix reports the responses. Accounting for the intensity and accumulation of past reforms confirms the baseline findings, with the responses being generally more precisely estimated, although smaller in magnitude.

Lag order and frequency. Figure B.16 examines the sensitivity of the estimates to the number of lags included in the local projections, considering specifications with between two and eight lags. The magnitude and timing of the responses remain broadly stable across specifications, indicating that the baseline results are not driven by the choice of four lags. Figure B.17 additionally reports estimates obtained using annual rather than quarterly data. Despite the lower frequency, the main response patterns remain unchanged: the gender employment gap narrows, female labor-force participation increases, and GDP rises over time. The response of the wage ratio is less clear-cut. As in the quarterly analysis, the effects are not immediate but emerge gradually, consistent with the anticipatory nature of the shock.

Subsamples and COVID period. The COVID pandemic generated large disruptions in labor-market and macroeconomic outcomes, and its inclusion in the sample may therefore affect the estimates. To address this concern, I augment the baseline specification with an indicator for the COVID period. Figure B.18 reports the results. The main response patterns remain broadly unchanged, indicating that the baseline findings are not driven by the pandemic period.

Figure B.19 examines the sensitivity of the results across alternative sample periods. The main response patterns remain broadly unchanged across subsamples. When the sample is restricted to the pre-COVID period, however, some responses are more muted.

5 Conclusion

This paper asks whether gender policies have consequences for the labor market and aggregate economic activity. To answer this question, I construct a narrative gender news series using the dates on which EU gender-equality directives are transposed into national law. The results show that gender-policy announcements increase female employment and labor-force participation, narrow the gender employment gap, and stimulate output over time. The rise in participation is preceded by an improvement in women's labor-market expectations, while the decline in wages points to an expansion in labor supply as an important adjustment mechanism.

These gains do not appear to crowd out male employment at the aggregate level, although the sectoral evidence points to a reallocation of male workers and broader downward pressure on wages. The effects also vary across policy objectives and implementation lags: work–family reconciliation policies generate particularly strong participation responses, while shorter transposition lags produce larger employment and wage effects. Overall, the results show that gender policies can have meaningful macroeconomic consequences. By focusing on EU directives, the analysis further shows that these policies generate real effects once incorporated into national law, and that timely transposition can strengthen their employment gains.

References

Acemoglu, D., Autor, D. H., and Lyle, D. (2004). Women, war, and wages: The effect of female labor supply on the wage structure at midcentury. *Journal of Political Economy*, 112(3):497–551.

Albanesi, S. (2025). Changing business cycles: The role of women's employment. *American Economic Journal: Macroeconomics*, 17(4):345–390.

Attanasio, O. P., Low, H., and Sánchez-Marcos, V. (2008). Explaining changes in female labor supply in a life-cycle model. *American Economic Review*, 98(4):1517–1552.

Baker, M., Halberstam, Y., Kroft, K., Mas, A., and Messacar, D. (2023). Pay transparency and the gender gap. *American Economic Journal: Applied Economics*, 15(2):157–183.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *The Quarterly Journal of Economics*, 131(4):1593–1636.

Barsky, R. B. and Sims, E. R. (2011). News shocks and business cycles. *Journal of Monetary Economics*, 58(3):273–289.

Beaudry, P. and Portier, F. (2006). Stock prices, news, and economic fluctuations. *American Economic Review*, 96(4):1293–1307.

Ben Zeev, N. and Pappa, E. (2017). Chronicle of a war foretold: The macroeconomic effects of anticipated defence spending shocks. *The Economic Journal*, 127(603):1568–1597.

Ben Zeev, N., Pappa, E., and Scola Gagliardi, E. (2025). The offensive power of defense news in europe. CEPR Discussion Paper 20637, CEPR Press, Paris & London.

Bergholt, D., Fosso, L., and Furlanetto, F. (2024). Macroeconomic effects of the gender revolution. Manuscript, Norges Bank.

Bertrand, M., Black, S. E., Jensen, S., and Lleras-Muney, A. (2019). Breaking the glass ceiling? the effect of board quotas on female labour market outcomes in norway. *The Review of Economic Studies*, 86(1):191–239.

Boneva, T., Golin, M., Kaufmann, K., and Rauh, C. (2026). Beliefs about maternal labour supply. *The Economic Journal*, 136(674):373–401.

Cloyne, J. (2013). Discretionary tax changes and the macroeconomy: new narrative evidence from the united kingdom. *American Economic Review*, 103(4):1507–1528.

Dagsvik, J. K., Kornstad, T., and Skjerpen, T. (2013). Labor force participation and the discouraged worker effect. *Empirical Economics*, 45(1):401–433.

den Besten, T., Barnichon, R., Känzig, D. R., and Singh, A. (2026). The macroeconomic effects of tariffs: Insights from 180 years of us trade policy. Technical report, National Bureau of Economic Research.

den Besten, T. and Känzig, D. R. (2026). Tariff changes in the u.s. from 1840–2024: A narrative analysis. Working paper.

Dube, A., Girardi, D., Jordà, Ò., and Taylor, A. M. (2025). A local projections approach to difference-in-differences. *Journal of Applied Econometrics*, 40(7):741–758.

Endo, H., Pallara, K., Sfregola, M., and Zanotti, L. (2026). When the treasury does monetary policy. *Available at SSRN 6312998*.

Farré, L., Felfe, C., González, L., and Schneider, P. (2026). Changing gender norms across generations: Evidence from a paternity leave reform. Technical report.

Fisher, J. D. and Peters, R. (2010). Using stock returns to identify government spending shocks. *The Economic Journal*, 120(544):414–436.

Fukui, M., Nakamura, E., and Steinsson, J. (2023). Women, wealth effects, and slow recoveries. *American Economic Journal: Macroeconomics*, 15(1):269–313.

Ghomi, M. and Pappa, E. (2026). Stimulating avenues: Eib loans and returns to public investment. CEPR Discussion Paper 21157, CEPR, Paris and London.

Goldberg, P. K., Gottlieb, C. T., Lall, S., Mehta, M., Peters, M., and Ratan, A. L. (2025). The global gender distortions index (ggdi). Technical report, National Bureau of Economic Research.

Greenwood, J., Seshadri, A., and Yorukoglu, M. (2005). Engines of liberation. *The Review of Economic Studies*, 72(1):109–133.

Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.

Hsieh, C.-T., Hurst, E., Jones, C. I., and Klenow, P. J. (2019). The allocation of talent and us economic growth. *Econometrica*, 87(5):1439–1474.

Hyland, M., Djankov, S., and Goldberg, P. K. (2020). Gendered laws and women in the workforce. *American Economic Review: Insights*, 2(4):475–490.

Jaimovich, N. and Rebelo, S. (2009). Can news about the future drive the business cycle? *American Economic Review*, 99(4):1097–1118.

Jordà, Ò. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

Känzig, D. R. (2021). The macroeconomic effects of oil supply news: Evidence from opec announcements. *American Economic Review*, 111(4):1092–1125.

Kleven, H., Landais, C., Posch, J., Steinhauer, A., and Zweimüller, J. (2024). Do family policies reduce gender inequality? evidence from 60 years of policy experimentation. *American Economic Journal: Economic Policy*, 16(2):110–149.

Lalive, R. and Zweimüller, J. (2009). How does parental leave affect fertility and return to work? evidence from two natural experiments. *The Quarterly Journal of Economics*, 124(3):1363–1402.

Maćkowiak, B. and Wiederholt, M. (2025). Rational inattention and the business cycle effects of productivity and news shocks. *American Economic Journal: Macroeconomics*, 17(1):274–309.

Mertens, K. and Ravn, M. O. (2012). Empirical evidence on the aggregate effects of anticipated and unanticipated us tax policy shocks. *American Economic Journal: Economic Policy*, 4(2):145–181.

Mertens, K. and Ravn, M. O. (2013). The dynamic effects of personal and corporate income tax changes in the united states. *American Economic Review*, 103(4):1212–1247.

Ngai, L. R., Olivetti, C., and Petrongolo, B. (2024). Gendered change: 150 years of transformation in us hours. Technical report, National Bureau of Economic Research.

Ngai, L. R. and Petrongolo, B. (2017). Gender gaps and the rise of the service economy. *American Economic Journal: Macroeconomics*, 9(4):1–44.

Olivetti, C., Pan, J., and Petrongolo, B. (2025). The evolution of gender in the labour market. VoxEU column.

Olivetti, C. and Petrongolo, B. (2017). The economic consequences of family policies: lessons from a century of legislation in high-income countries. *Journal of Economic Perspectives*, 31(1):205–230.

Ramey, V. A. (2011). Identifying government spending shocks: It's all in the timing. *The Quarterly Journal of Economics*, 126(1):1–50.

Ramey, V. A. and Zubairy, S. (2018). Government spending multipliers in good times and in bad: evidence from us historical data. *Journal of Political Economy*, 126(2):850–901.

Reiche, L. and Maffei-Faccioli, N. (2026). Divergent perceptions, divergent pay: Inflation and the gender wage gap. CESifo Working Paper 12835, CESifo, Munich.

Romer, C. and Romer, D. (2004). A new measure of monetary shocks: Derivation and implications. *American Economic Review*, 94(4):1055–1084.

Romer, C. D. and Romer, D. H. (2010). The macroeconomic effects of tax changes: estimates based on a new measure of fiscal shocks. *American Economic Review*, 100(3):763–801.

Ruhm, C. J. (1998). The economic consequences of parental leave mandates: Lessons from europe. *The Quarterly Journal of Economics*, 113(1):285–317.

Schmitt-Grohé, S. and Uribe, M. (2012). What's news in business cycles. *Econometrica*, 80(6):2733–2764.

A Data

Table 4 gives details on the data sources, variable definitions, and transformations used in the analysis.

Table 4: Data description, sources, and transformations

Variable	Description	Source	Transformation
Economic policy uncertainty	Country-specific index of economic policy uncertainty.	Baker et al. (2016)	Level
Employment ratio	Female-to-male employment ratio (%).	ILOSTAT	100×log ratio
Female employment[18]	Number of employed women aged 15–64.	ILOSTAT	100 × log
Female hours worked	Average hours worked by employed women aged 15-64.	ILOSTAT	100 × log
Female labor-force participation	Labor-force participation among women aged 15–64.	OECD	100 × log
Female part-time employment	Number of women aged 15-64 employed part-time.	ILOSTAT	100 × log
Female unemployment expectations	Women's expectations about future aggregate unemployment.	ECB CES	Level
Female wages	Average hourly wage of female employees.	ILOSTAT	100 × log
Gender equality[19]	Women, Business and the Law Index.	World Bank	Level
Government expenditure	Final consumption expenditure of general government.	Eurostat	100 × log

Continued on next page

[18]For variables disaggregated by sex and sector, the definitions, sources, and transformations are the same as for the corresponding baseline variables; only the sex and sector considered differ. For brevity, the table reports only the baseline variables. The same applies to variables disaggregated by women's age, marital status, or educational attainment.

[19]This variable is used in the maximum-share approach.

Table 4 continued

Variable	Description	Source	Transformation
Inflation	Quarterly inflation based on the Harmonised Index of Consumer Prices.	Eurostat	$100 \times \Delta \log$
Job vacancies	Number of job vacancies (thousands).	Eurostat	Level
Male employment	Number of employed men aged 15–64.	ILOSTAT	$100 \times \log$
Male unemployment	Number of unemployed men aged 15–64.	ILOSTAT	$100 \times \log$
Male unemployment expectations	Men's expectations about future aggregate unemployment.	ECB CES	Level
Male wages	Average hourly wage of male employees.	ILOSTAT	$100 \times \log$
Misallocation	Global Gender Distortion Index.	Goldberg et al. (2025)	Level
National transposition	Indicator equal to one in a quarter in which at least one national transposition measure is adopted and zero otherwise.	EUR-Lex	Level
Real GDP	Real gross domestic product.	Eurostat	$100 \times \log$
Total unemployment	Number of unemployed people aged 15–64.	ILOSTAT	$100 \times \log$
Wage ratio	Female-to-male wage ratio (%).	ILOSTAT	$100 \times$log ratio

Additional notes: Before applying the transformations reported in the final column, a centered moving average is computed separately for each country to seasonally adjust the series. Since seasonal adjustment may not be innocuous, I assess their robustness to alternative methods. The corresponding results are reported in Figure B.20.

B Additional results

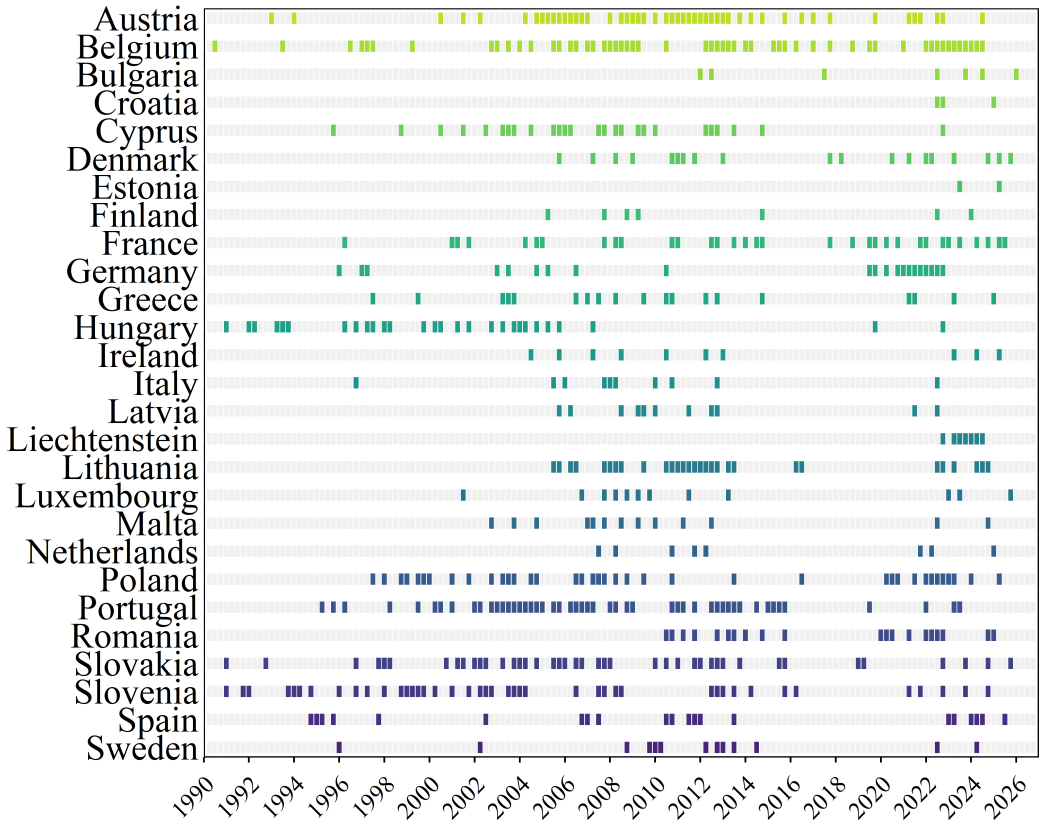


Figure B.1: Gender news series by country

Notes: Each row represents a country and each cell a quarter. Colored cells identify quarters in which the gender-policy news indicator equals one, corresponding to the national transposition of an EU gender-equality directive.

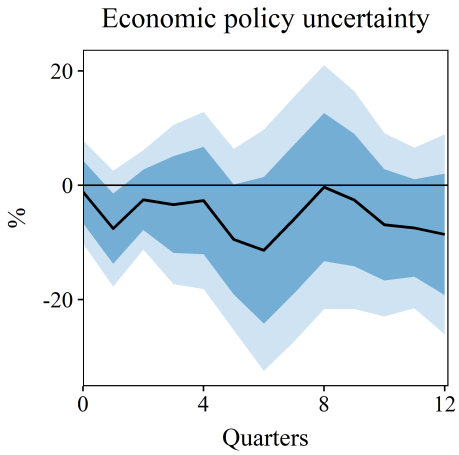


Figure B.2: Impulse responses of economic policy uncertainty after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

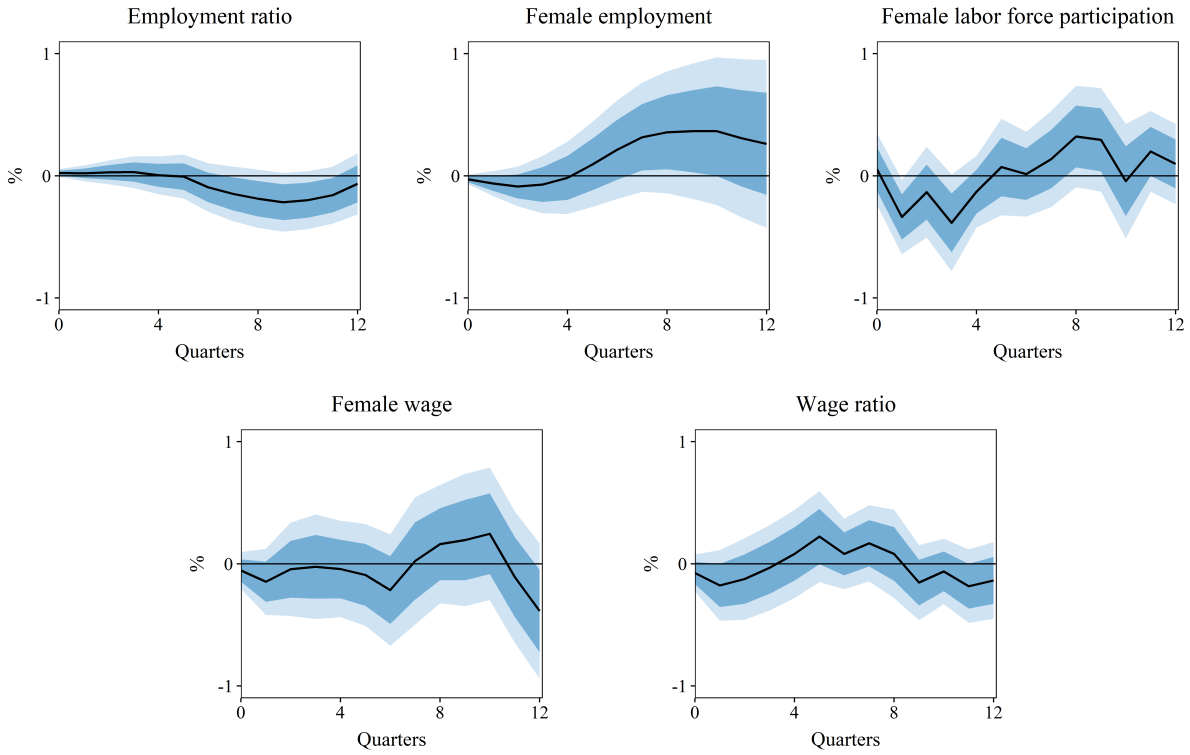


Figure B.3: Impulse responses to a placebo shock

Notes: Impulse responses to a unit placebo shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

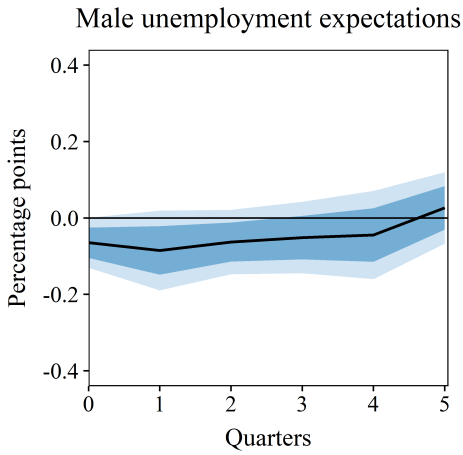


Figure B.4: Response of male unemployment expectations to a gender news shock

Notes: Impulse response of male unemployment expectations to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate; the dark and light shaded areas show the 68 and 90 percent confidence bands, respectively.

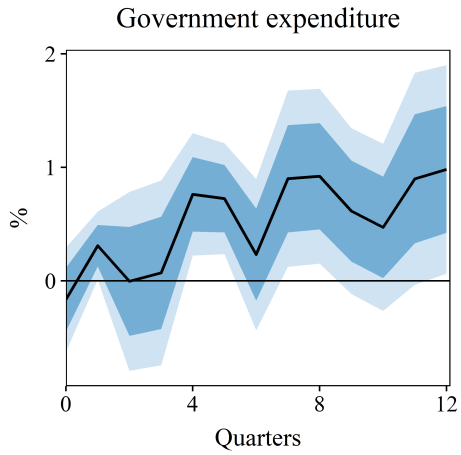


Figure B.5: Impulse responses of government expenditure after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively. The specification does not control for GDP.

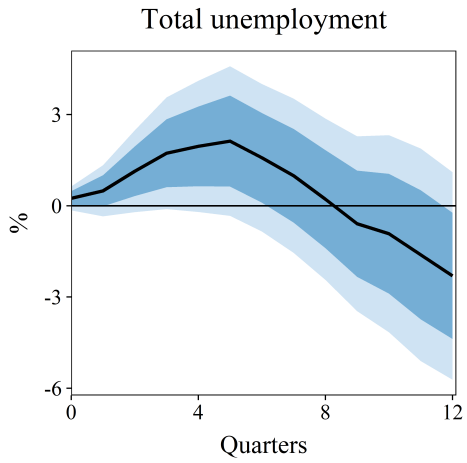


Figure B.6: Impulse responses of total unemployment after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.



Figure B.7: Impulse responses of male unemployment after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.



Figure B.8: Impulse responses of part-time female employment after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

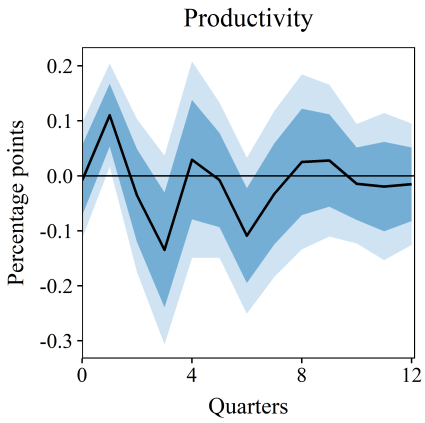


Figure B.9: Impulse responses of productivity after a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

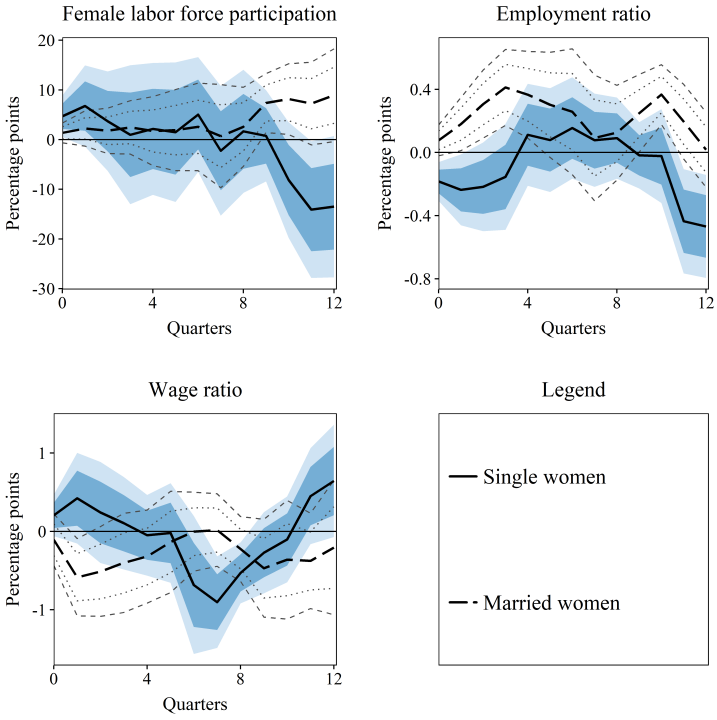


Figure B.10: Impulse responses to a gender news shock, by marital status

Notes: The figure reports impulse responses to a unit gender news shock, corresponding to one national transposition, separately for single and married women. The estimation sample includes France only. Solid lines denote the responses of single women, while dashed lines denote the responses of married women.

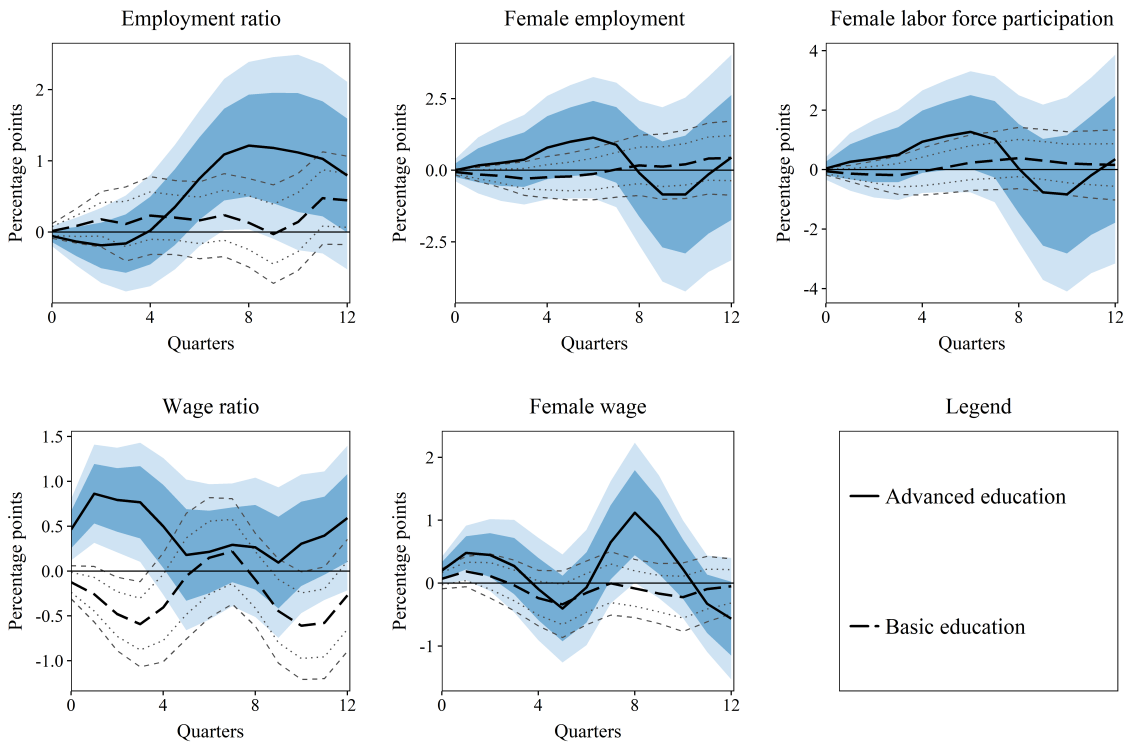


Figure B.11: Impulse responses to a gender news shock, by educational attainment

Notes: The figure reports impulse responses to a unit gender news shock, corresponding to one national transposition, separately for women with advanced and basic education. Solid lines denote the responses of women with advanced education, while dashed lines denote the responses of women with basic education. Shaded areas indicate 68 and 90 percent confidence bands.

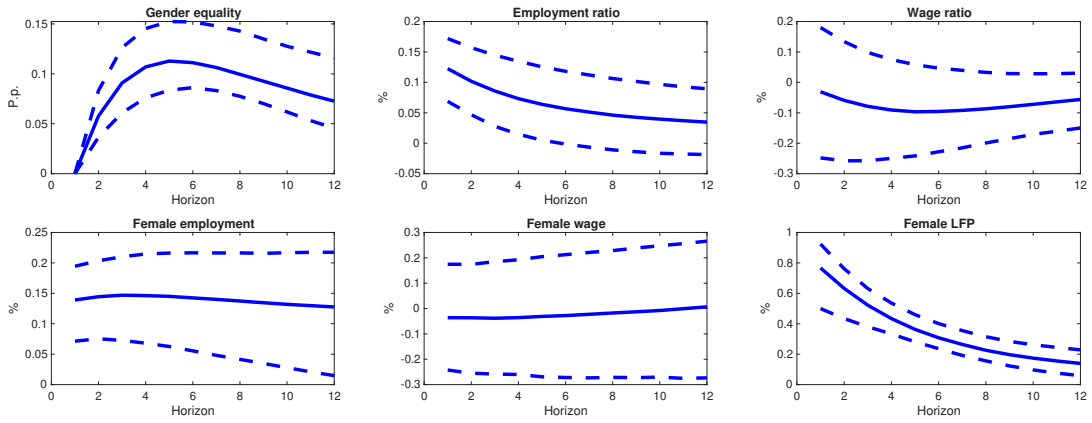


Figure B.12: Impulse responses after a gender news shock

Notes: Impulse responses of gender news shock identified with maximum forecast error variance approach. The solid line is the point estimate and the dashed lines are 90 percent credible bands.

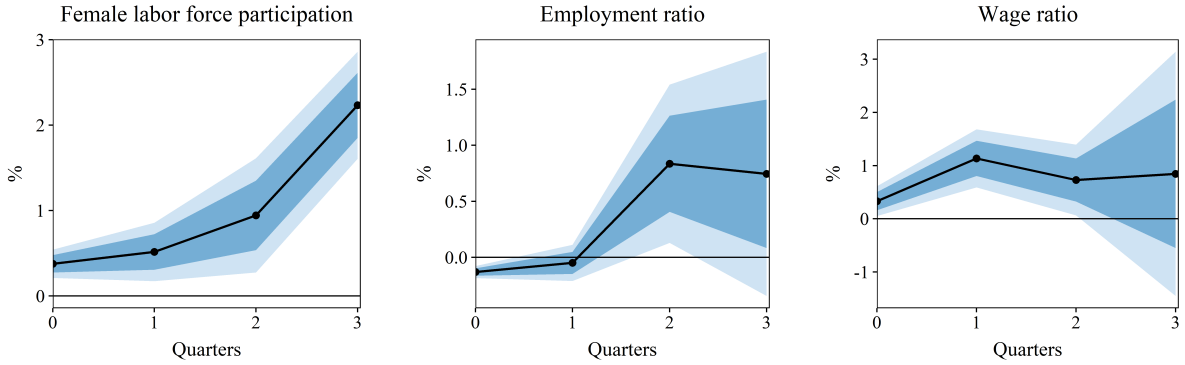


Figure B.13: Impulse responses to a gender news shock

Notes: The figures report lp-DiD estimates of the effects of one national transposition. The solid lines show the point estimates; the dark and light shaded areas represent the 68 and 90 percent confidence bands, respectively.

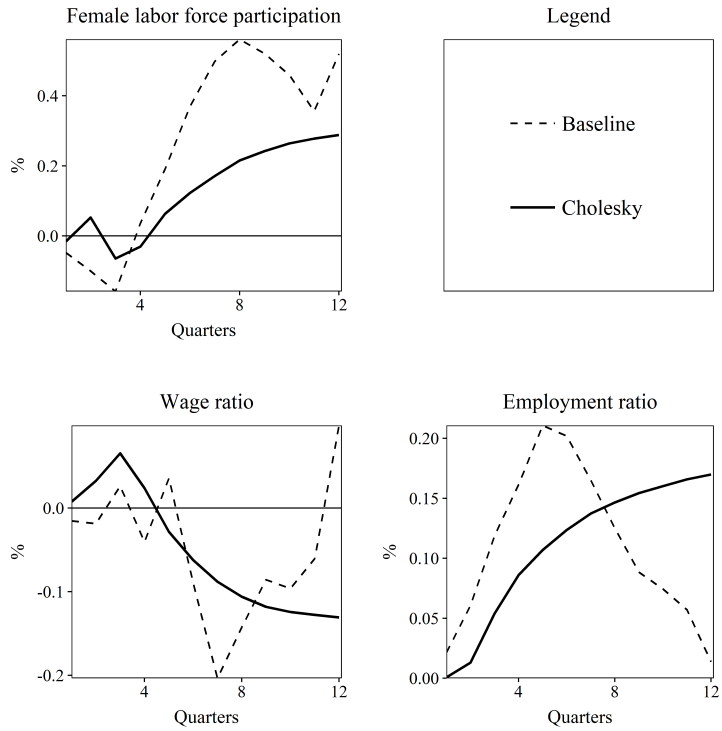


Figure B.14: Impulse responses: baseline versus recursive VAR

Notes: The dashed black line reports the baseline estimates; the solid black line reports the Cholesky estimates.

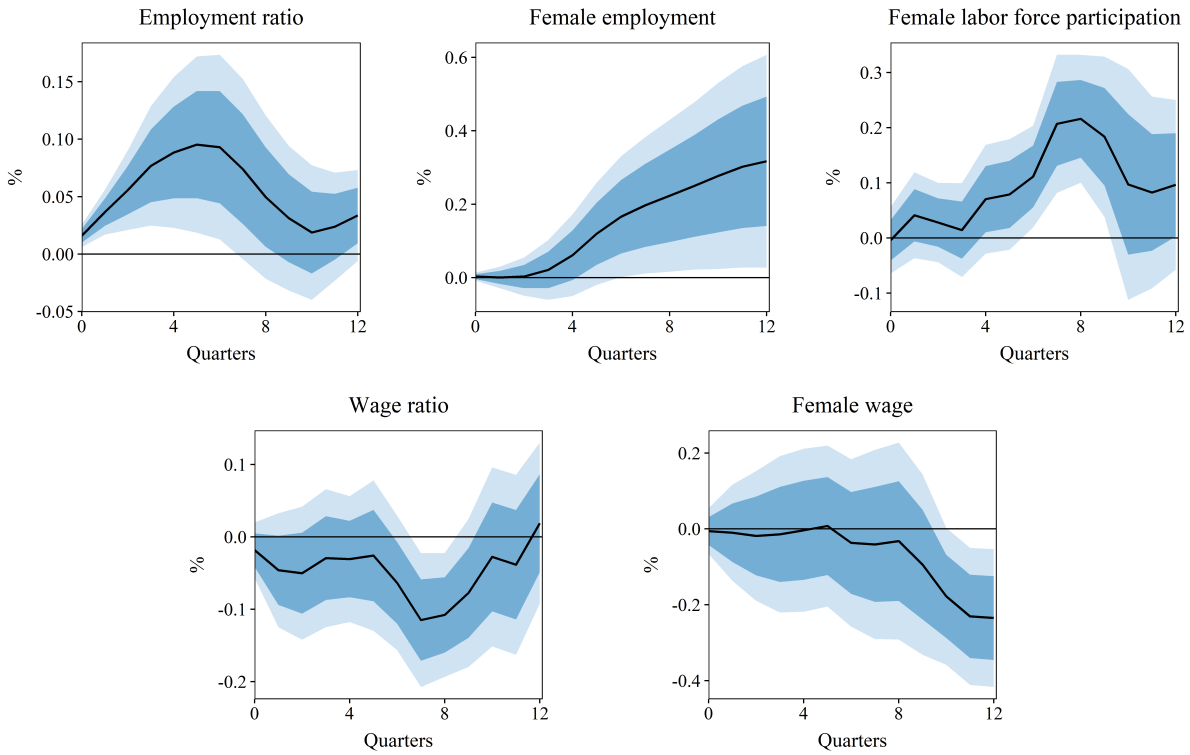


Figure B.15: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock (controlling for intensity), corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are 68 and 90 percent confidence bands, respectively.

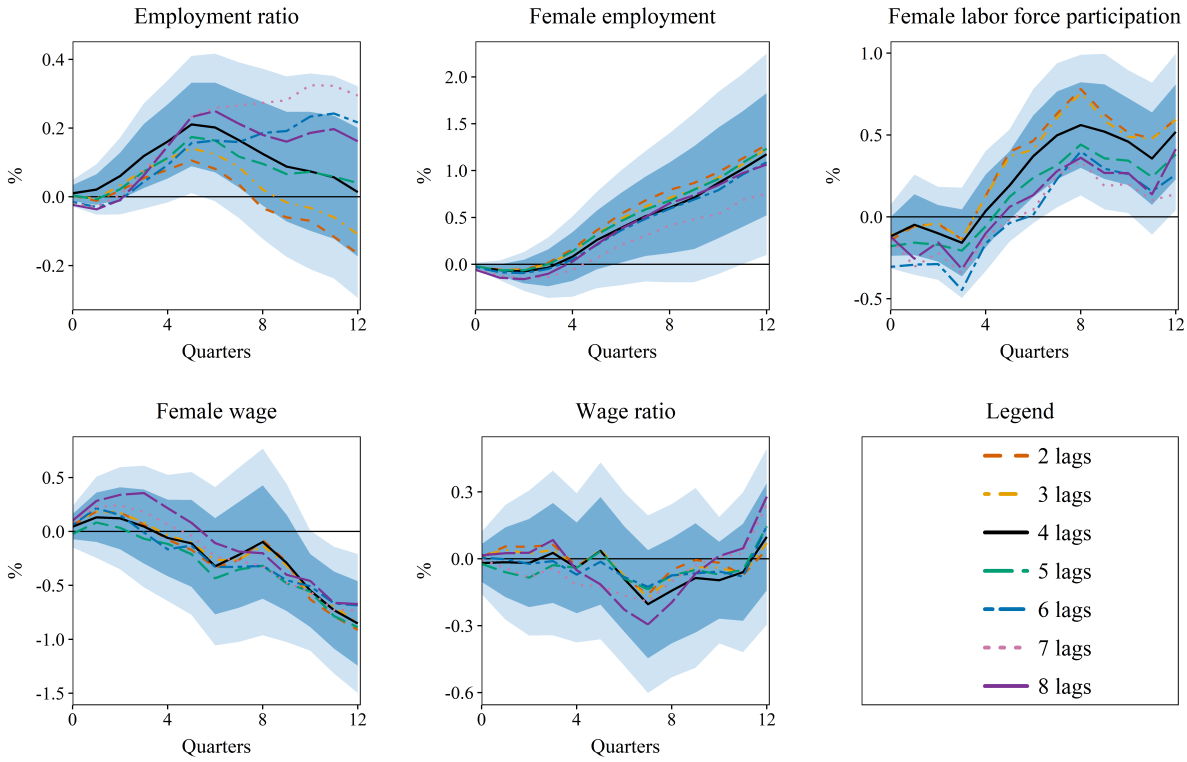


Figure B.16: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender-policy news shock, corresponding to one national transposition. The solid line is the baseline point estimate and the dark and light shaded areas are baseline 68 and 90 percent confidence bands, respectively. The dashed lines represent different lag orders.

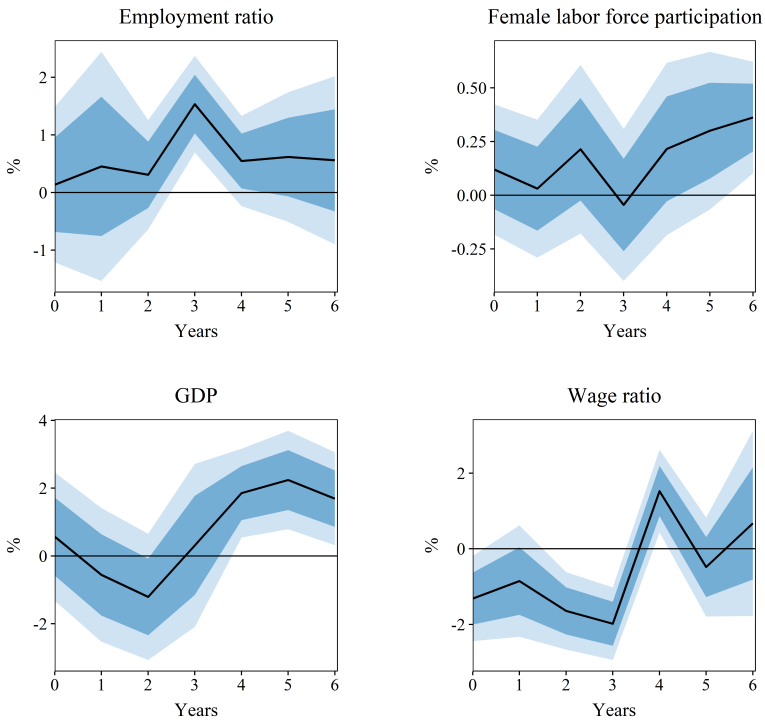


Figure B.17: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the point estimate and the dark and light shaded areas are baseline 68 and 90 percent confidence bands, respectively.

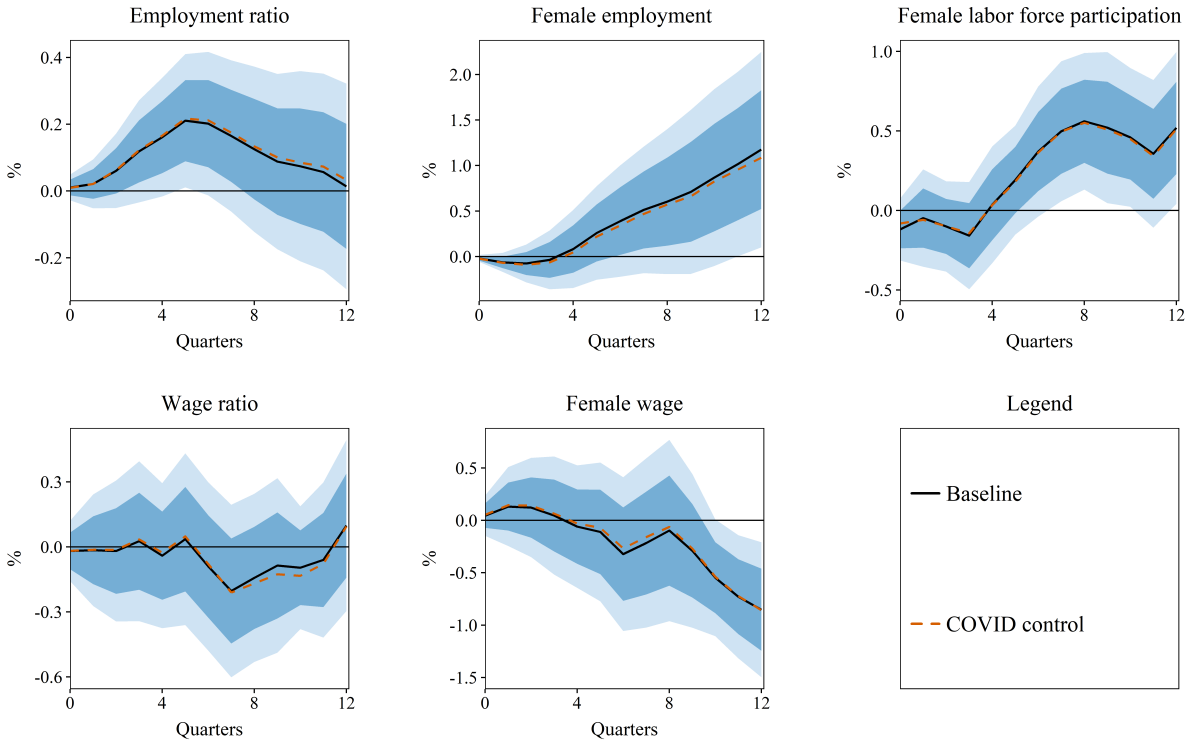


Figure B.18: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the baseline point estimate and the dark and light shaded areas are baseline 68 and 90 percent confidence bands, respectively.

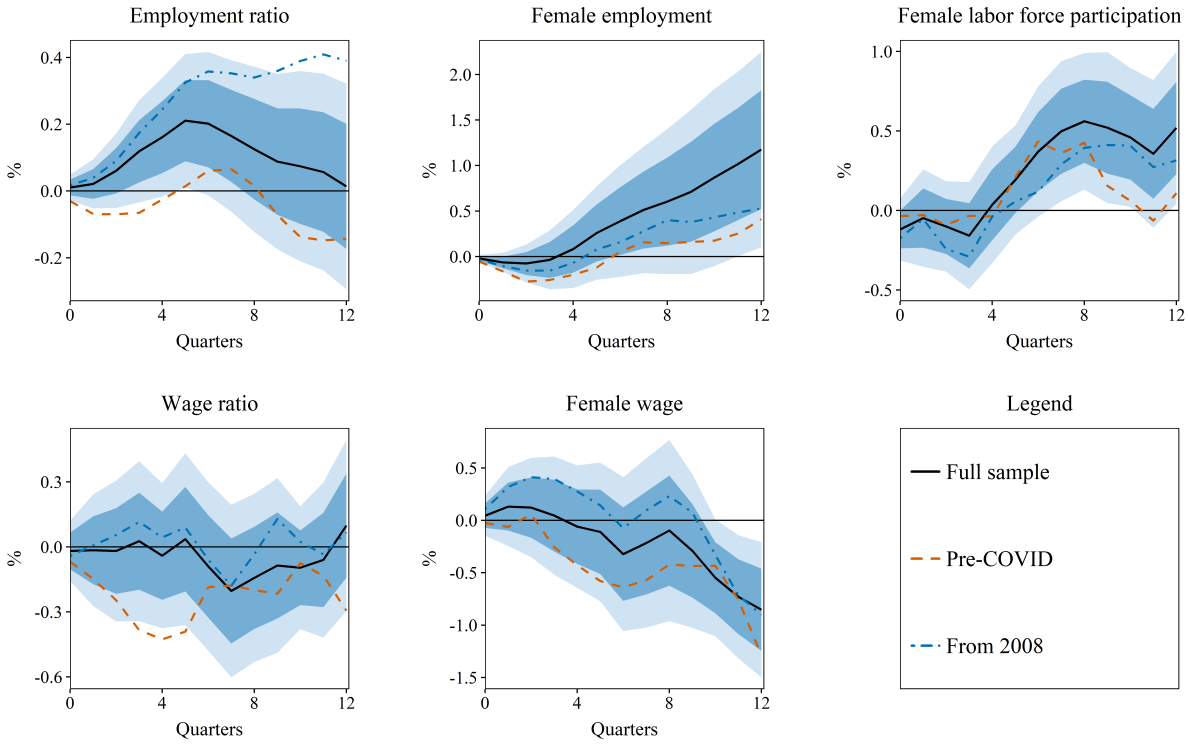


Figure B.19: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The solid line is the baseline point estimate and the dark and light shaded areas are baseline 68 and 90 percent confidence bands, respectively. The dashed lines represent different sample periods.

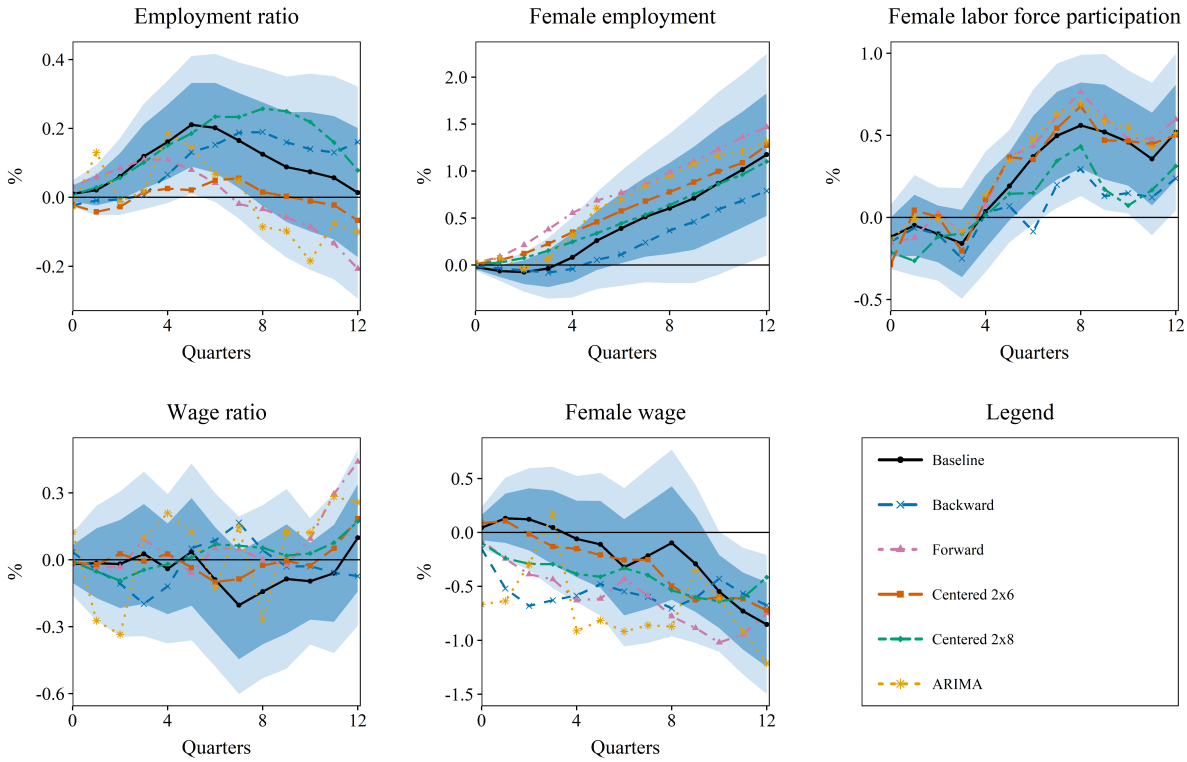


Figure B.20: Impulse responses to a gender news shock

Notes: Impulse responses to a unit gender news shock, corresponding to one national transposition. The figure compares different seasonal adjustments. The shaded areas show the baseline 68 and 90 percent confidence bands.